MODELING GAS-LIQUID FLOW IN PIPES: FLOW PATTERN TRANSITIONS AND DRIFT-FLUX MODELING

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By

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I certify that I have read this report and that in my opinion it is fully adequate, in scope and in quality, as partial fulfillment of the degree of Master of Science in Petroleum Engineering.

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Abstract

Two-phase gas-liquid flow in pipes is of great practical importance in petroleum engineering. This work focuses on the determination of flow pattern transitions and drift-flux modeling in gas-liquid flow. Using the data in the Stanford Multiphase Flow Database as well as other data from the literature, we investigate transition predictions in mechanistic models and the use of the drift-flux model for holdup calculations. The flow pattern prediction in the Petalas & Aziz (1998) mechanistic model is evaluated. Other transition criteria are also compared with experimental data. Barnea's (1986) model is shown to give the best results for prediction of the transition to dispersed bubble flow. It is demonstrated that this transition in the Petalas & Aziz (1998) mechanistic model can be improved by tuning a parameter used in their model. Approximation of the interfacial friction factor in stratified flow via the gas/wall friction factor is recommended for use in the transition predictions from stratified flow. For the transition to annular-mist flow, a holdup based transition criterion is shown to give reasonable results. The effects of fluid properties on flow pattern transitions are also presented using the data of Weisman *et al.* (1979). Fluid properties are shown to have less effect on flow pattern transitions than the inclination angle of the pipe.

Use of the drift-flux model over multiple flow patterns is investigated next. Using the in situ gas volume fraction to represent the flow pattern information, we fit the drift-flux model parameters C_0 (distribution parameter) and V_d (drift velocity) as linear functions of α_G (in situ gas volume fraction). The method proposed in this work is shown to provide much better gas volume fraction predictions than previous methods. However, a general correlation is not given in this work, since the resulting correlation for V_d is not entirely consistent with expected behavior at high α_G. Approximate results for the effects of inclination angle on V_d are also presented. Using our current data (measured in pipe diameters of 1-2 inches), the drift-flux model used in Eclipse is evaluated. A modification

of the user-definable parameters in the model is suggested to improve the performance of the Eclipse model at high $\alpha_{G.}$

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Using the software *Digitizer* in GIS laboratory, I digitized the flow pattern transition data of Shoham (1982), Weisman *et al.* (1979), and Kokal & Stainslav (1987). Otherwise, the evaluation of transition models in this work would have been much harder.

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Chapter 1

1. Introduction

1.1. Overview

The two-phase flow of gas and liquids has many applications in the petroleum industry. It is commonly encountered in the production and transportation of oil and gas. For example, the oil that flows to the surface is often accompanied by gas. Pipeline flow may also contain two or more flowing phases.

The complexity in the prediction and design of gas-liquid systems lies in the simultaneous existence of the gas and liquid phases. The interface between the two phases can be distributed in many configurations. This phenomenon is called flow pattern, which is a very important feature of two-phase flows. In single-phase flow in pipes, the design parameters such as pressure drop can be calculated in a relatively straightforward way. However, the existence of a second phase presents difficult challenge in the understanding and modeling of the flow system. The hydrodynamics of the flow, as well as the flow mechanisms, change significantly from one flow pattern to another. For instance, it has been demonstrated (Cheremisinoff, 1986) that for similar flow conditions, slug flow and wavy flow may result in a difference in pressure drop of a factor of two. Some heat transfer parameters estimated using the stratified flow correlations might change by several orders of magnitude from those estimated by the annular flow correlations.

In petroleum engineering applications, the three most important hydrodynamic features are the flow pattern, the holdup of the two phases, and the pressure drop. In order to estimate accurately the pressure drop and holdup, it is necessary to know the actual flow pattern under the specific flow conditions. However, the procedure for determining flow patterns is nontrivial. Further, the discontinuities in pressure drop and holdup due to the

shift from one flow regime to another may give rise to convergence problems when a wellbore flow model is coupled with a reservoir simulator. Thus, some simplified models with the underlying flow pattern information incorporated are also required.

1.2. Literature Review

Experimental work plays a very important role in multiphase flow research. Due to the complexity of the problem itself, the experiments provide us with the most direct and reliable way of understanding the physical mechanisms. Based on the data from these experiments, various models can be developed and the accuracy of these different models can be examined. However, the improved understanding of multiphase flow in pipes requires a combined experimental and theoretical approach (Brill & Arirachakaran, 1992). In the solution of engineering problems, there are several levels of approaches (Taitel, 1995): empirical correlations, modeling techniques and rigorous solution of Navier-Stokes equations. We now consider each of these approaches.

Historically, empirical correlation is a very useful engineering approach, and a large number of correlations appear in the literature. Although some of them are very widely used in the oil and gas industry, empirical correlations are generally valid only for the parameter ranges for which they are generated.

Another possibility is the use of Computational Fluid Dynamics for the calculation of pressure drop and volume fractions in gas-liquid pipe flows. This approach is in principle applicable to a wider range of applications. However this procedure calls for a solution of the continuity, momentum and energy equations for the two fluids and the determination of the gas-liquid interface. In addition, the well-posedness and stability of the problem is still open to question (Taitel *et al.*, 1989). To date, only some simplified calculations have been performed. Arif (1999) calculated the pressure drop for single phase flow in a wellbore with radial influx. Newton & Behnia (2000) performed calculations for the stratified gas-liquid flow, and demonstrated results in agreement with those from the mechanistic model of Taitel & Dukler (1976).

Modeling techniques lie between the empirical correlations and numerical solution of the Navier-Stokes equations. They approximate the problem at hand by considering the most important physical phenomenon, while neglecting the less important effects which may complicate the problem but do not improve significantly the accuracy of the solution. This is probably the most appropriate approach from an engineering perspective --- the problem is approximated and formulated in a way that it can be analyzed with reasonable effort. In this work, we will focus on the modeling of gas-liquid flows. This will include mechanistic modeling of flow pattern transitions and drift-flux modeling for holdup calculations.

1.2.1. Mechanistic Model

Mechanistic modeling started with the work by Taitel *et al.* (1976, 1980). It took into account the physical mechanism behind the transitions to different flow patterns. Although their work only considered the transitions among different flow patterns (the calculation of pressure drop and holdup was not included), it was the pioneering work along these lines and opened the door for improved models for each flow pattern. Barnea (1987) presented a unified model valid for the whole range of pipe inclination angles, which enabled various models to be linked together through her unified flow pattern transition criteria.

Following the work by Taitel and Barnea, comprehensive mechanistic models have been presented by Xiao *et al.* (1990), Ansari *et al.* (1994), Kaya *et al.* (1999), Gomez *et al.* (2000) and Petalas & Aziz (1997, 1998). These models contain the determination of flow patterns and the computation of pressure drop and hold up. The transition criteria in the Petalas & Aziz (1998) model are based on Barnea's work. In addition, based on the data in the Stanford Multiphase Flow Database, some new correlations are developed for liquid/wall and liquid/gas interfacial friction in stratified flow and for the liquid fraction entrained and the interfacial friction in annular-mist flow (Petalas & Aziz, 1998).

Flow pattern depends on the phase flow rates, fluid properties of both phases, pipe diameter and pipe inclination angle. To investigate what happens in upward and

downward flow not only helps us understand the underlying mechanism of flow pattern transitions, but also contributes to the development of prediction tools, since in practice, wellbores and pipelines can be vertical, horizontal or deviated. The effect of pipe inclination on the flow pattern transition in gas-liquid flow has been studied both experimentally and theoretically (Shoham 1982). Barnea's model was developed based on Shoham's data, and it has been tested against experimental data over the entire range of pipe inclinations. For the effects of fluid properties, Weisman *et al.* (1979) conducted experiments in horizontal pipes. However, the ability of mechanistic models to capture the effects of fluid properties has not been analyzed extensively.

There are several transitions among the major flow patterns, and sub-regime transitions exist within some of the main flow patterns. In many cases, specific physical mechanisms can be associated with these transitions. In horizontal and slightly inclined pipes, the transition from stratified flow is based on a Kelvin-Helmholtz instability analysis on the wave growth in the liquid surface (Taitel & Dukler, 1976). This transition criterion is relatively well-established, and was further adjusted by Barnea (1987) to handle the stratified flow transition in downward flow. For other transitions, various transition mechanisms and subsequent models have been proposed; e.g., the analysis of buoyant forces and forces due to turbulent fluctuations (Taitel & Dukler (1976) and Kokal & Stanislav (1987)) for the transition to dispersed bubble flow. Also for the transition to annular-mist flow, the spontaneous blockage of the gas core (Barnea, 1986) and the effective viscosity criterion (Joseph *et al.*, 1996) have been suggested.

1.2.2. Drift-Flux Model

As indicated above, the mechanistic model is not suitable for all applications. Mechanistic models work most efficiently where the flow pattern information cannot be ignored. The choice of the model depends on the application and on the time and cost constraints. In transient gas-liquid flow simulation, or in the coupled simulation of reservoir and wellbore flow, the flow model is required to be simple, continuous and differentiable (Schlumberger GeoQuest, 2000). Mechanistic models are not suitable for this purpose so some simplified models are needed. The drift flux model is one such

model. With two parameters (distribution/profile parameter C_0 and drift velocity V_d), holdup can be calculated from the superficial velocities.

The drift flux model was first proposed by Zuber & Findlay (1965). Usually, this model is applied to vertical dispersed systems. In the Petalas & Aziz (1997, 1998) mechanistic model, in the intermittent flow, dispersed bubble flow and bubble flow regimes, the holdup is calculated by the drift flux model and then the pressure drop is obtained using a homogeneous model. Some effort has also been put into the investigation of the two parameters C_0 and V_d. Petalas & Aziz (1997) correlated the profile parameter and drift velocity with the liquid Reynolds number using the data points in the above three flow patterns. Mishima & Ishii (1984) related the profile parameter with fluid densities. This expression was used by Ouyang (1998) in a homogeneous model with slip for gas-liquid wellbore flow. Similarly, Eclipse (Schlumberger GeoQuest, 2000) uses the drift flux model in calculations for the multi-segment wells. The correlations in Eclipse were synthesized from several other published correlations. They depend on fluid properties, gas volume fraction, mixture velocity and pipe inclination angle.

1.3. Proposed Work

This work consists of two parts --- modeling of flow pattern transition in mechanistic models and applying the drift-flux model in holdup calculations. This includes:

Evaluate the flow pattern predictions in the Petalas and Aziz mechanistic model (1998). This model has been tested extensively with respect to the calculation of pressure drop and holdup (Petalas & Aziz, 1997, 1998). However, the comparison of flow patterns between the experimental data and the proposed model is not extensive. They report only the fraction of points for which the model correctly predicts the flow pattern.

Investigate the effects of fluid properties on flow pattern transitions. The experimental data by Weisman *et al.* (1979) will be used to test the ability of the Petalas and Aziz mechanistic model (1998) to capture the impact of fluid properties.

Evaluate the performance of newly published transition models. Though there are a large number of models for several transitions, we will focus on transitions to dispersed bubble flow and annular mist flow. The comparisons of the different transition models against the experimental data will be presented.

Analyze the steady state holdup data in the Stanford Multiphase Flow Database. We will apply the drift-flux model to different flow patterns. A method will be proposed to develop correlations between the drift flux model parameters and the gas volume fractions. Using available data, the drift-flux correlations in Eclipse will be evaluated and tuned.

1.4. Report Outline

This report begins with a discussion of some basic concepts in two-phase flow, including classification of the different flow patterns and the descriptions of flow pattern maps. The prediction of flow pattern transitions by the Petalas & Aziz (1998) mechanistic model is presented in Chapter 3. The effect of fluid properties on flow pattern transitions is illustrated. Different transition criteria for transitions to dispersed bubble flow and to annular mist flow are evaluated. Chapter 4 describes the drift flux model used in holdup calculations. Our work on the determination of drift flux model parameters is then presented. Chapter 5 contains a summary of this work and some suggestions for future research.

Chapter 2

2. Basic Concepts in Two-Phase Gas Liquid Flows

In this chapter, some basic concepts and variables describing the gas-liquid flow system are presented and discussed. The flow patterns encountered in horizontal and vertical pipe flows are described. Flow pattern maps are also introduced as a means to represent the flow pattern information.

2.1. Definition of Basic Parameters

The superficial velocities of the liquid and gas phases (V_{SL} and V_{SG}) are defined as the volumetric flow rate for the phase divided by the pipe cross sectional area:

$$V_{SL} = \frac{Q_L}{A} \quad and \quad V_{SG} = \frac{Q_G}{A}, \tag{2-1}$$

where Q_L and Q_G are the volumetric flow rate of liquid and gas respectively and A is the pipe cross sectional area.

The mixture velocity is given by the sum of the gas and liquid superficial velocities:

$$V_M = V_{SL} + V_{SG}. \tag{2-2}$$

The input volume fractions of the liquid and gas phases (C_L and C_G) are defined as:

$$C_L = \frac{Q_L}{Q_L + Q_G} = \frac{V_{SL}}{V_M}, \tag{2-3}$$

$$C_G = \frac{Q_G}{Q_L + Q_G} = \frac{V_{SG}}{V_M}. \tag{2-4}$$

By definition the sum of the liquid and gas volume fractions is equal to one.

The characteristic of two-phase flow is the simultaneous flow of two phases of different density and viscosity. Usually in horizontal and uphill flows, the less dense and/or less viscous phase tends to flow at a faster velocity. In gas-liquid flow, gas moves much faster than liquid except in downward flow. The difference in the in situ average velocities between the two phases results in a very important phenomenon --- the "slip" of one phase relative to the other, or the "holdup" of one phase relative to the other (Govier & Aziz, 1972). This makes the in situ volume fractions different than the input volume fractions. Although "holdup" can be defined as the fraction of the pipe volume occupied by a given phase, holdup is usually defined as the in situ liquid volume fraction, while the term "void fraction" is used for the in situ gas volume fraction.

Let the cross sectional area occupied by liquid be A_L ; the remaining area A_G is occupied by gas. The liquid holdup and gas volume fraction are defined as:

$$\alpha_L = \frac{A_L}{A} \quad and \quad \alpha_G = \frac{A_G}{A}. \tag{2-5}$$

After the in situ volume fraction is known, we can calculate the average (in situ) velocity for each phase:

$$V_L = \frac{Q_L}{A_L} = \frac{V_{SL}}{\alpha_L}, \tag{2-6}$$

$$V_G = \frac{Q_G}{A_G} = \frac{V_{SG}}{\alpha_G}. \tag{2-7}$$

These are the true average velocities of liquid and gas phases, which are larger than the superficial velocities.

Fluid properties (density, viscosity and interfacial tension) for each phase and geometric parameters such as the pipe internal diameter and pipe inclination angle also have an influence on the performance of the system. In this work, the pipe inclination angle θ is measured from the horizontal except when otherwise noted.

2.2. Flow Patterns

In gas-liquid flow, the interface between the two phases can exist in a wide variety of forms, depending on the flow rate, fluid properties of the phases and the geometry of the system. Flow patterns are used to describe this distribution. Hubbard & Dukler (1966) suggested three basic flow patterns: separated, intermittent and distributed flow.

- Separated flow patterns: Both phases are continuous. Some droplets or bubbles of one phase in the other may or may not exist. Separated flow patterns include:
  - Stratified flows: Stratified smooth flow and stratified wavy flow.
  - Annular flows: Annular film flow and annular-mist flow, which entrains liquid droplets in the gas core.

- Intermittent flow patterns: At least one phase is discontinous. These flow regimes include:
  - Elongated bubble flow.
  - Slug flow, plug flow.
  - Churn or froth flow (a transition zone between slug flow and annular-mist flow).

- Dispersed flow patterns: In these flow regimes, the liquid phase is continuous, while the gas phase is discontinous. Flow patterns include:
  - Bubble flow.
  - Dispersed bubble flow, in which the finely dispersed bubbles exist in a continuous flowing liquid phase.

We will describe in detail the features of these flow patterns for both horizontal and vertical flows.

2.2.1. Flow Patterns in Horizontal Pipes

In Fig. 2-1, the flow patterns observed in horizontal pipes are illustrated schematically:

- In stratified flow, the gas and liquid flow separately with the liquid phase in the lower portion of the pipe. The stratified flow pattern is subdivided into stratified smooth flow, where the liquid surface is smooth, and stratified wavy flow where the interface is wavy. The stratified smooth flow takes place in low liquid and gas flow rates. As the gas rate increases, instability of the liquid surface results in the occurrence of stratified wavy flow.

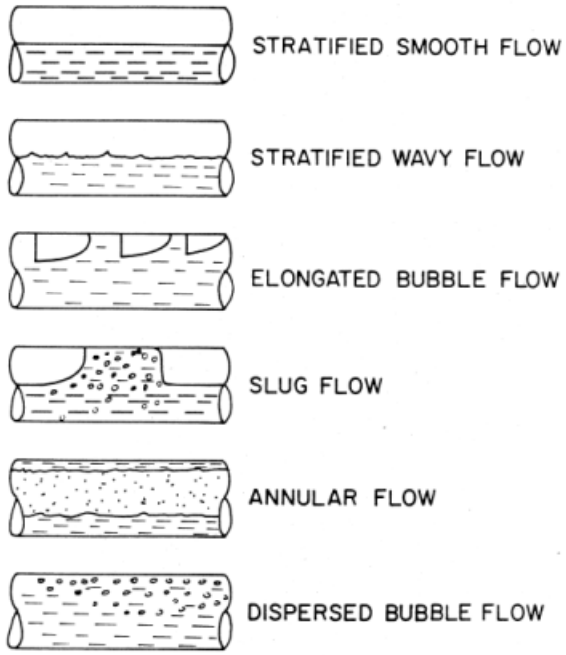


Figure 2-1: Schematic of flow patterns in horizontal pipes (from Shoham, 1982)

- Intermittent flow patterns are characterized by the alternate appearance of slugs and gas bubbles in the pipes. The major difference between elongated bubble flow and slug flow is that in elongated bubble flow there are no entrained gas bubbles in the liquid slugs.

- When gas rates increase, annular (also referred to as annular-mist) flow occurs. The liquid flows as a film around the pipe wall and a gas core forms in the middle. The gas core may contain some entrained liquid droplets. In this flow pattern, the gas rate

needs to be high enough to support the gas core in the middle and prevent the liquid film from falling down.

- Unlike annular-mist flow, dispersed bubble flow usually occurs at high liquid flow rates. The liquid phase is continuous while the gas phase is distributed as discrete bubbles.

2.2.2. Flow Patterns in Vertical Pipes

Fig. 2-2 illustrates the flow patterns observed in vertical flow:

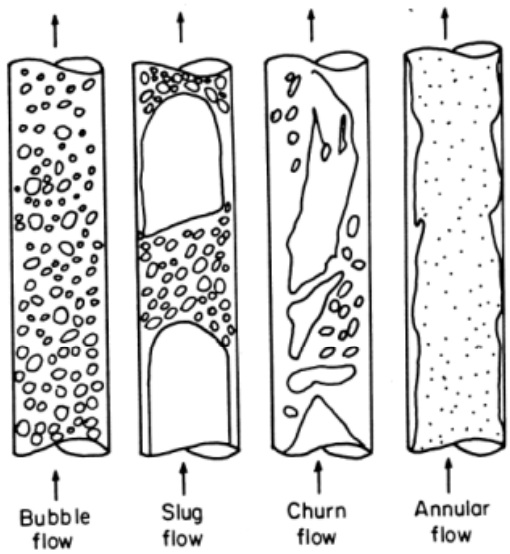


Figure 2-2: Schematic of flow patterns in vertical flow (from Shoham, 1982)

- At low liquid velocities, the gas is dispersed as discrete bubbles. This flow regime is called bubble flow. As the liquid flow rate increases, the bubbles may increase in size via coalescence. Generally, the gas phase is dispersed as discrete bubbles in the liquid continuum. The distinction between bubbly and dispersed bubble flow is not clearly visible (Barnea, 1987). The bubbly flow pattern is observed only in vertical and off-vertical flows in relatively large diameter pipes, while dispersed bubble flow is normally found over the whole range of pipe inclinations.

- From bubble flow, with a further increase in gas flow rate, some of the bubbles coalesce to form larger, longer, cap-shaped bubbles. These large bubbles are termed Taylor bubbles. Slug flow consists of Taylor bubbles, separated by regions of bubbly flow called slugs. A thin liquid film flows downwards around Taylor bubbles. The distribution of Taylor bubbles in vertical flow is symmetric.

- In churn flow (also called froth flow), the bubbles and the slugs become highly distorted and appear to merge at high gas flow rates. Another difference between slug flow and churn flow is that the falling film of the liquid surrounding the gas plugs cannot be observed in churn flow.

- Similar to the annular-mist flow in a horizontal pipe, the annular flow here is characterized by the liquid flowing as a film around the pipe wall, surrounding a high velocity gas core, which may contain entrained liquid droplets. The upward flow of the liquid film against gravity results from the forces exerted by the fast moving gas core.

2.2.3. Observations of Flow Patterns in Inclined Pipes

Pipe inclination angles have a very strong influence on flow pattern transitions. Shoham (1982) experimentally showed that in the transition from stratified flow to non-stratified flow, even a small change in the angle has a major effect. Deviations from the horizontal tend to diminish the separation between the gas and the liquid phases. In practice, stratified flow is not observed in the experimental range of flow rates for upward inclinations higher than about 20°. For downward flow, however, the stratified flow region is commonly observed up to -70°.

2.3. Flow Pattern Maps

For a given system, with Q_L and Q_G specified, a particular flow pattern will result. Flow pattern is often displayed using a flow pattern map, which is a two-dimensional map depicting flow regime transition boundaries. The selection of appropriate coordinates to present clearly and effectively the different flow regimes has been a research topic for a

long time. Although dimensionless variables are preferred in theory, the dimensional coordinates such as superficial velocities are much more generally used in practice. Actually, we show later in Chapter 4 that we can use other variables to present clearly the flow pattern information, such as the mixture velocity and the gas volume fraction. But, since the volume fraction is usually unknown (and it is one of our objectives to determine volume fractions based on the flow pattern information), this is not always a practical way to present transition boundaries.

The generation of flow pattern maps falls into two categories. One is the experimental flow pattern map generated directly from experimental data. Fig. 2-3 illustrates a very commonly used experimental flow pattern map, which was generated from a large amount of experimental data. It is completely empirical and limited to the data on which it is based. To account for the effects of fluid properties and pipe diameter, additional correlations must be introduced.

Mechanistic flow pattern maps, by contrast, are developed from the analysis of physical transition mechanisms, which are modeled by fundamental equations. In the literature, various transition mechanisms have been proposed. In Chapter 3, some of them will be analyzed and evaluated. In these transition models, the effects of system parameters are incorporated, so they can be applied over a range of conditions (one example is shown in Fig. 2-4). One thing we must point out here is that empirical correlations are still required in the mechanistic model for the model closure.

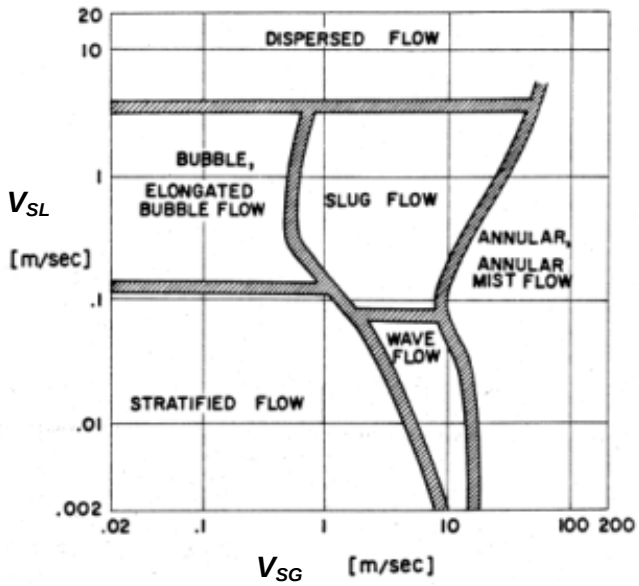


Figure 2-3: Experimental flow pattern map (Mandhane *et al.* (1974), air-water system, horizontal pipe)

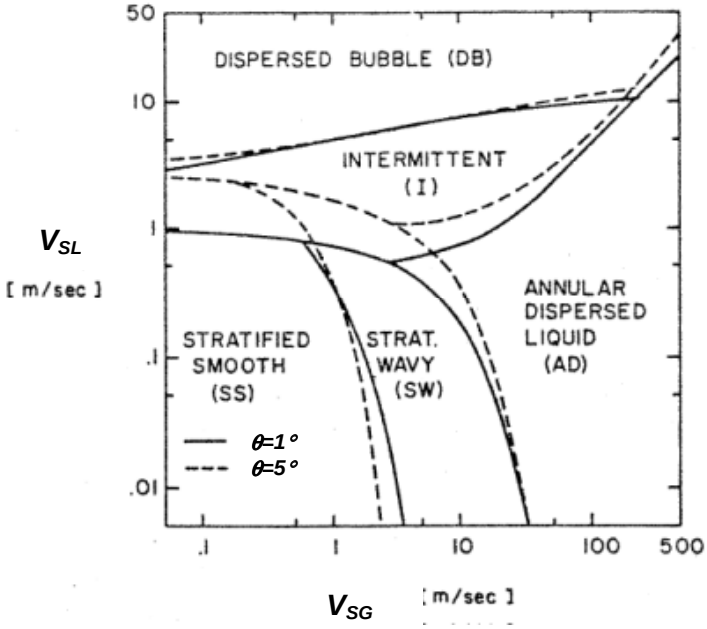


Figure 2-4: Mechanistic flow pattern map (Taitel *et al.* (1976), air-water system, slightly downward pipe)

Chapter 3

3. Flow Pattern Transitions in Mechanistic Models

The generation of flow pattern maps by mechanistic models is considered in this chapter. We first present the overall performance of the transition predictions in the Petalas & Aziz (1998) mechanistic model for horizontal flow. Effects of fluid properties are illustrated using the experimental data of Weisman *et al.* (1979). Then, some existing transition models for dispersed bubble and annular-mist flows are evaluated. The interfacial friction factor in stratified flow, which affects the transition between stratified and intermittent flows, is also discussed.

3.1. Transition Predictions of the Petalas & Aziz (1998) Mechanistic Model

The Petalas & Aziz (1998) mechanistic model includes flow pattern predictions and calculations for pressure drop and holdup. The transition model is based on the unified model for the whole range of pipe inclinations proposed by Barnea (1987). Predictions for pressure drop and holdup have undergone extensive testing using the data in the Stanford Multiphase Flow Database (SMFD) and have proven to be more accurate than other existing models (Petalas & Aziz, 1998). For detailed descriptions of the model development and implementation, refer to Petalas & Aziz (1997, 1998). In this section, we will evaluate the performance of the model for transition predictions in horizontal flow.

In order to assess a model, accurate and consistent data is required. However, the flow pattern transition data may display consistency problems, since a subjective interpretation is often involved in labelling a flow pattern. Before applying the data, a consistency check was performed. More detailed descriptions of the data and the consistency checks are given in Appendix A.

In the flow pattern map shown in Fig. 3-1, we use data points with different colors to represent the various flow patterns observed in the experiment. This map closely matches the empirical flow pattern map (Fig. 2-3) of Mandhane *et al.* (1974). All the flow patterns illustrated schematically in Fig. 2-1 appear in Fig. 3-1. The major transitions are the transition to dispersed bubble flow, the transition to annular-mist flow and that between intermittent flow and stratified flow. For comparison, we also display the transition boundaries given by the Petalas & Aziz (1998) mechanistic model in Fig. 3-1. Overall, it gives us fairly good predictions, especially in the transition to annular-mist flow from either intermittent flow or stratified flow, and in the transition from stratified smooth to stratified wavy flow. The major problem lies in the transition to the dispersed bubble flow and that between the intermittent flow and stratified flow, both of which are overestimated by the model.

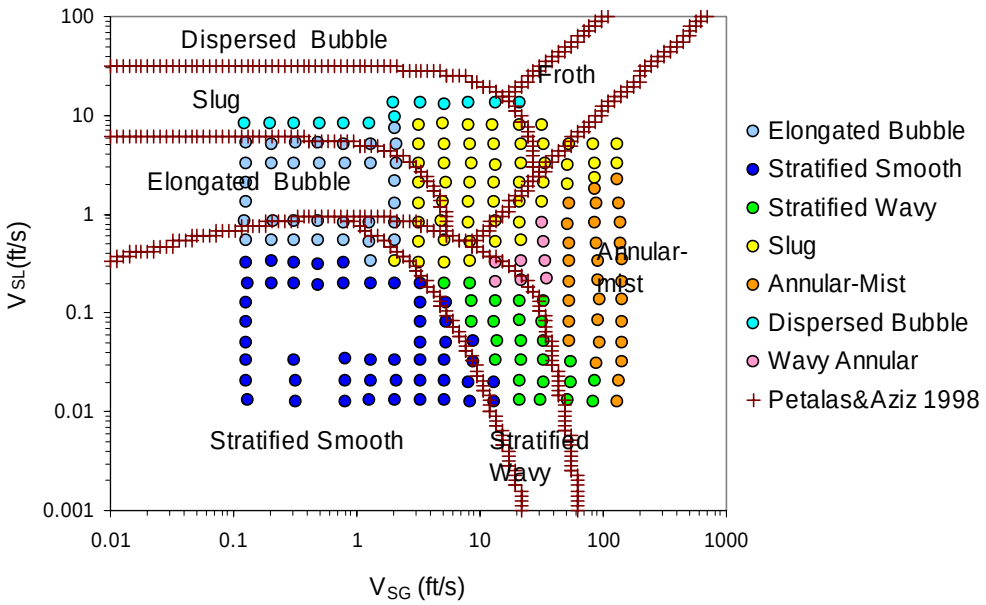


Figure 3-1: Comparison of transition boundaries (Data: Shoham (1982), air-water system, horizontal flow, *D*=1.0 inch)

Similar results are shown in Fig. 3-2. The data here is from the same researcher (Shoham, 1982), but with a pipe diameter of 2 inch, rather than 1 inch as in Fig. 3-1. Similar observations are again obtained. The overestimation of the transition between intermittent

flow and stratified flow is more obvious in this case. For the case D=2.0 inch, we also show the comparisons with other data in Figs. 3-3 and 3-4. All these data indicate that dispersed bubble flow occurs at a liquid flow rate of about 10 ft/s or less. However, the model prediction for this transition is at V_{SL}=30 ft/s. The experimental observation for the transition between intermittent flow and stratified flow is at V_{SL}=0.2~0.5 ft/s, while the model predicts that it takes place at a significantly higher liquid flow rate.

These comparisons indicate that the current transition model can be improved. Later in this chapter, we will evaluate some other transition criteria together with those currently used in the Petalas & Aziz (1998) mechanistic model. Discussion of transitions at other inclination angles will also be included.

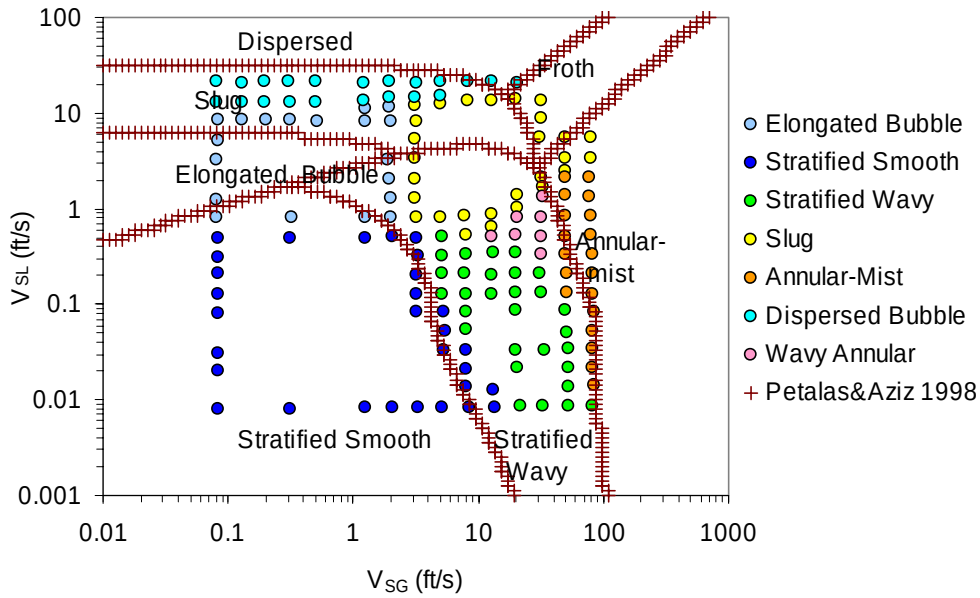


Figure 3-2: Comparison of transition boundaries (Data: Shoham (1982), air-water system, horizontal flow, D=2.0 inch)

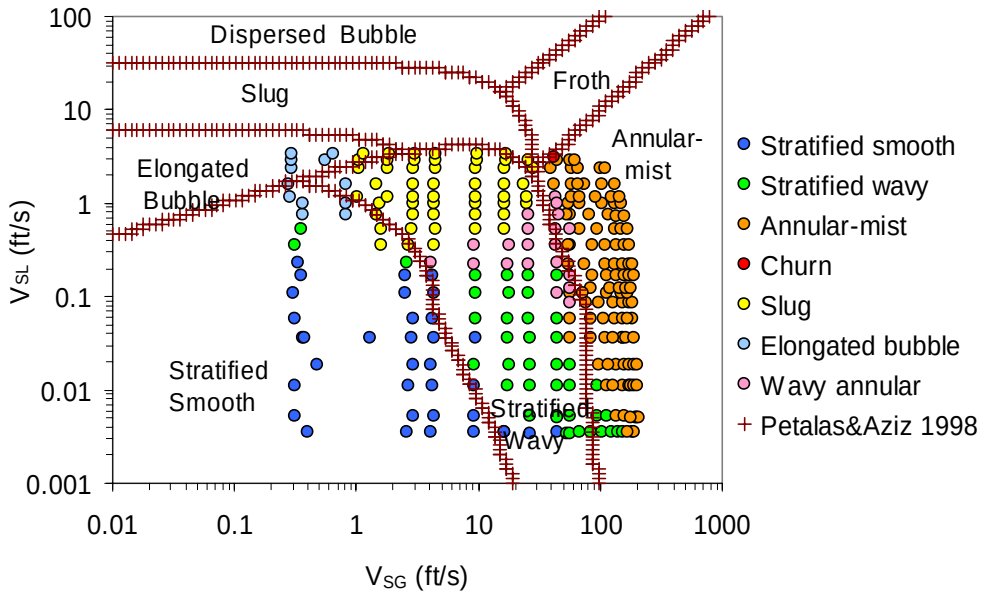


Figure 3-3: Comparison of transition boundaries (Data: Spedding & Nguyen (1976), air-water system, horizontal flow, *D*=1.79 inch)

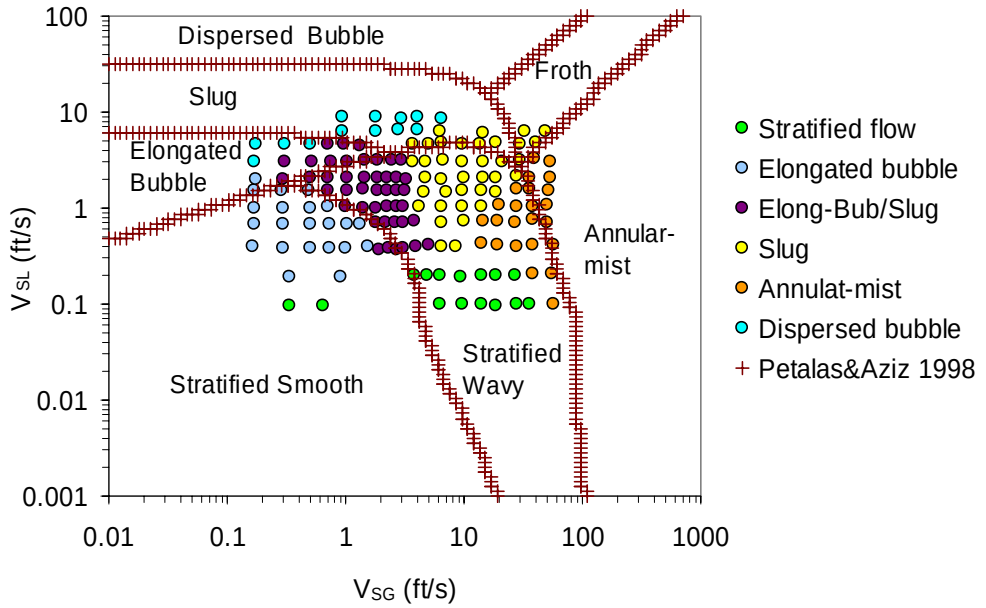


Figure 3-4: Comparison of transition boundaries (Data: Kokal & Stanislay (1987), air-water system, horizontal flow, *D*=2.02 inch)

3.2. Effects of Fluid Properties

We know that flow patterns depend on pipe inclinations and fluid properties. Compared to the investigation of inclination angles, the effects of fluid properties have received less attention. However, the study of flow pattern transitions for various fluid properties is helpful to our understanding of the physical transition mechanisms. Weisman *et al.* (1979) carried out experiments in horizontal pipes to investigate the influence of fluid properties on flow pattern transitions. Using their experimental data, we will now evaluate the Petalas & Aziz (1998) mechanistic model with regard to its ability to capture the impact of changing fluid properties.

Table 3-1 presents a summary of the experimental data of Weisman *et al.* (1979). Their study of liquid viscosity and interfacial tension is with a 2 inch pipe, while that on vapor density is with a 1 inch pipe. Although the fluids were selected to allow large changes in one property while having relatively insignificant changes in the other properties, we see that the resulting parameters for the case of vapor density are still not ideal --- a significant reduction of interfacial tension is also observed.

Table 3-1: Summary of Experimental Data of Weisman *et al.* (1979)

	Base Case	Effect of Liquid Viscosity	Effect of Interfacial Tension	Effect of Vapor Density
ρ_L *(lb/ft^3)*	62.4	77.4	62.4	84.3
ρ_G *(lb/ft^3)*	0.08	0.08	0.08	**2.7**
μ_L *(cp)*	1.0	**150.0**	1.0	0.3
σ *(dyne/cm)*	65	65	**38**	9.5

Note: Horizontal pipe, μ_G = 0.01cp.

The effect of liquid viscosity on flow pattern transition is shown in Fig. 3-5. The map for the increased liquid viscosity shows relatively little change from that obtained with the air-water system (Fig. 3-5(a)), even though the liquid viscosity is varied from 1.0 cp to

150.0 cp. Of the major transition boundaries, only the transition to dispersed bubble flow is shifted slightly to lower liquid flows. Fig. 3-5(b) displays the corresponding results obtained by the Petalas & Aziz (1998) mechanistic model. The transition to stratified flow moves to significantly lower liquid flow rates, which is not indicated by the experimental data.

Fig. 3-6 demonstrates the influence of interfacial tension, which is reduced by half from 65 dyne/cm to 38 dyne/cm. From the experimental data (Fig. 3-6(a)), we see that the transition to annular and dispersed flow and the transition between intermittent and separated flow are essentially unchanged. The major change observed is the sub-regime transition within stratified flow and intermittent flow. The stratified wavy/stratified smooth flow transition occurs at higher gas flow rates. The predictions by the mechanistic model are displayed in Fig. 3-6(b). According to the model, interfacial tension has no impact at all on the flow pattern transitions. We know, however, that the major difference between stratified smooth flow and stratified wavy flow is the shape of the interface between the liquid and gas phases. Therefore, we expect the interfacial tension to have some impact on this transition, as indicated by the experimental data in Fig. 3-6(a).

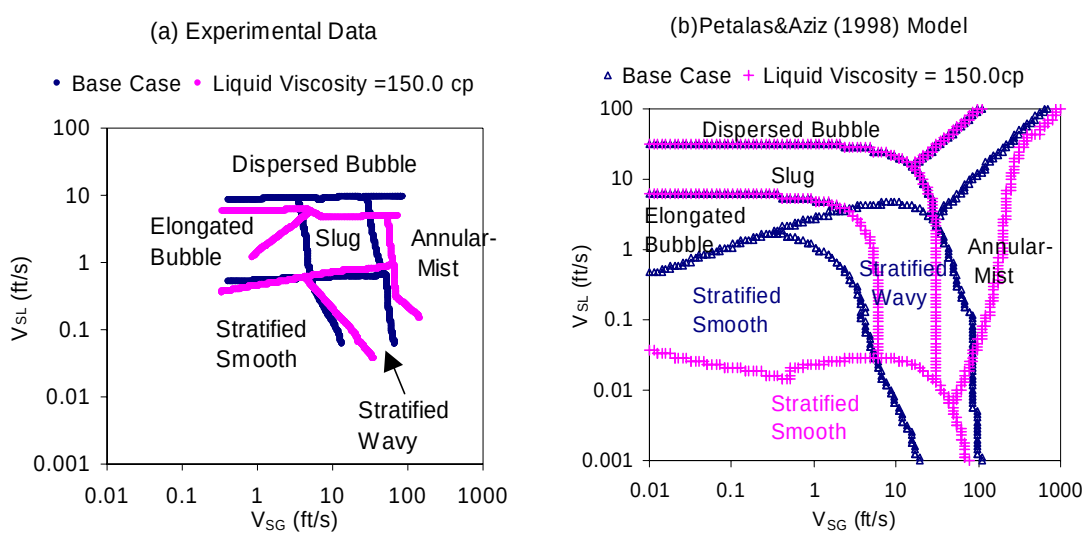


Figure 3-5: Effects of liquid viscosity (Data: Weisman *et al.* (1979), horizontal flow, *D*=2 inch)

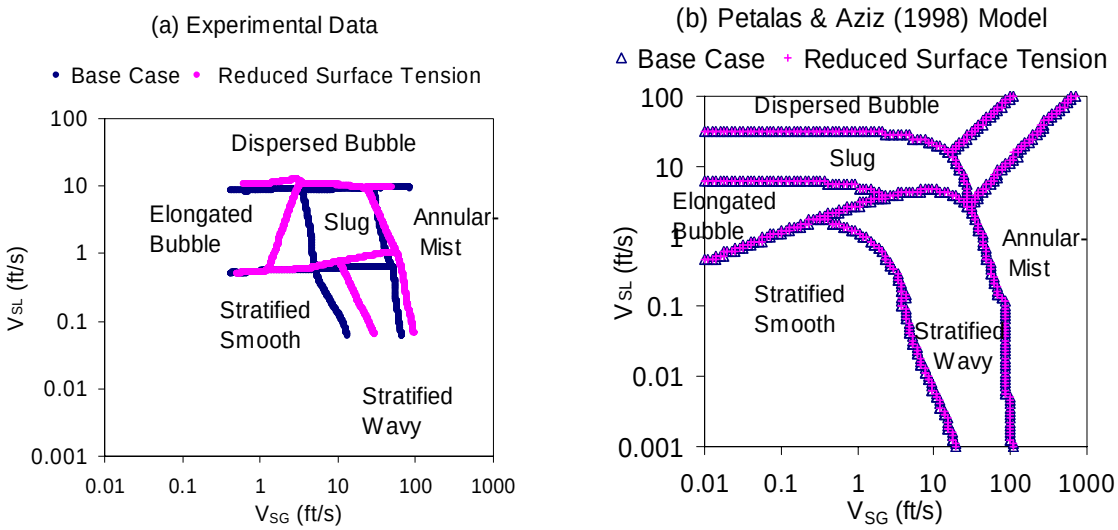


Figure 3-6: Effects of surface tension (Data: Weisman *et al.* (1979), horizontal flow, *D*=2 inch)

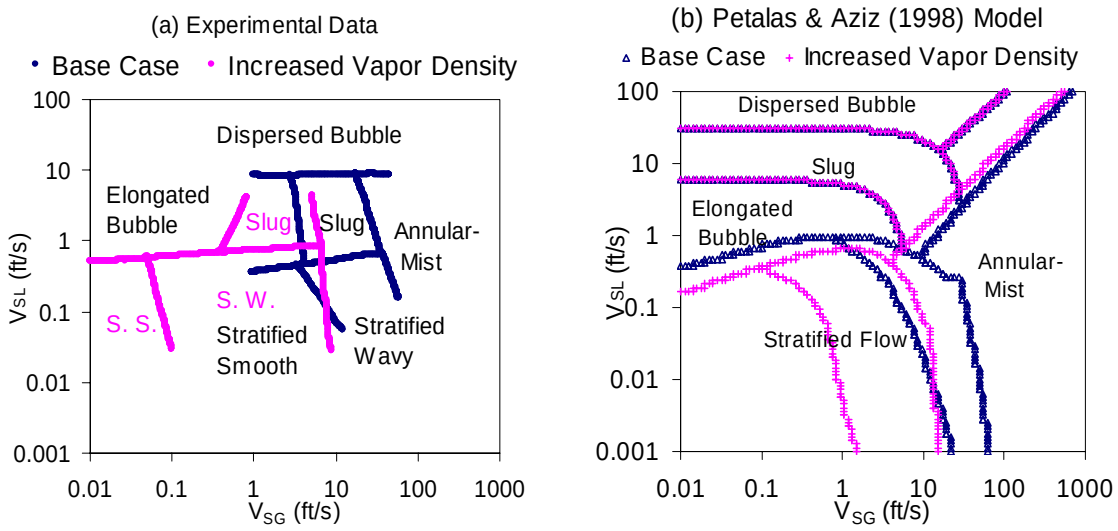


Figure 3-7: Effects of vapor density (Data: Weisman *et al.* (1979), horizontal flow, *D*=1 inch)

The effect of liquid vapor density is examined next (Fig. 3-7). Recall that in this case, the surface tension is also changed significantly (Table 3-1). Our observation from Fig. 3-6(a), which shows that surface tension has a minimal effect on transitions, suggests that the major change in Fig. 3-7(a) is due to the liquid vapor density itself. We see the transition to annular flow and that between stratified smooth flow and stratified wavy flow occurs at much lower gas flow rates. The explanation for this could be that the

lighter the gas phase, the higher the gas flow rate needed to support the gas core in the middle at the pipe. This trend is also observed in the mechanistic model results in Fig. 3-7(b).

Based on the experimental data of Weisman *et al.* (1979), we can draw the following conclusions. Compared with the effects of inclination angles, fluid properties appear to have less impact on flow patterns. Vapor density has a greater effect than other fluid properties. The Petalas & Aziz (1998) mechanistic model gives the correct trend for stratified/annular-mist and stratified smooth/wavy transitions. The impact of surface tension is not represented in the current mechanistic model.

3.3. Evaluation of Other Transition Criteria

Flow pattern predictions rely on the transition criteria, and also the correlations used. The existing models for transitions to dispersed bubble flow and to annular-mist flow will be evaluated, and the correlation for interfacial friction factors in stratified flow will be shown to have a strong impact on the transition prediction between stratified and intermittent flows.

3.3.1. Transition to Dispersed Bubble Flow

Dispersed bubble flow is observed at high liquid flow rate and low gas flow rate. Usually, turbulent forces due to the high liquid flow rate are considered to play an important role in the break up of gas bubbles. Based on this mechanism, several transition models have been proposed. We will first briefly describe these models and then compare them with experimental observations.

- Models for horizontal or near-horizontal flows

1) Taitel & Dukler (1976):

The transition was considered in stratified flow. The gas phase is at the top of the pipe due to buoyant forces. The transition to dispersed bubble flow takes place when the turbulent fluctuations overcome the buoyant forces so that the gas tends to mix with the

liquid. The buoyant force (F_B) and turbulent force (F_T) were evaluated per unit length of the gas region:

$$F_B = g(\cos\theta)(\rho_L - \rho_G)A_G\,, \tag{3-1}$$

where A_G is the gas cross sectional area, and

$$F_T = \frac{1}{2}\rho_L \overline{v'^2} S_i = \frac{1}{2}\rho_L (V_L^2 \frac{f_{wL}}{2}) S_i\,, \tag{3-2}$$

where S_i is the interfacial perimeter, v' the fluctuation part of turbulent velocity, which was approximated by Taitel & Dukler (1976) using the average liquid velocity and the liquid/wall friction factor f_{wL}. When $F_T \geq F_B$, that is

$$V_L \geq \left[\frac{4A_G}{S_i}\frac{g\cos\theta}{f_{wL}}\left(1-\frac{\rho_G}{\rho_L}\right)\right]^{\frac{1}{2}}\,, \tag{3-3}$$

dispersed bubble flow occurs. Here, we should point out that since the forces are analyzed in the geometry of stratified flow, V_L is computed from the momentum equations in stratified flow, as are the geometric parameters A_G and S_i.

2) Kokal & Stanislav (1987):

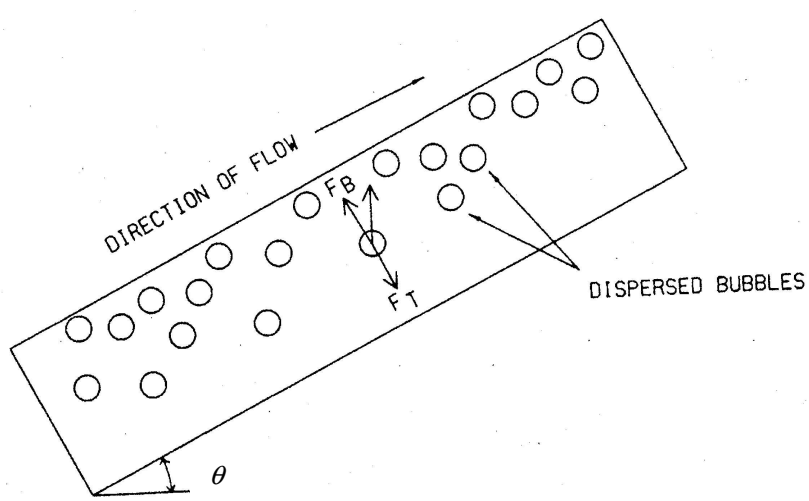


Figure 3-8: Analysis of forces in dispersed bubble flow (from Kokal & Stainslav, 1987)

Kokal & Stanislav (1987) modified the model by Taitel & Dukler (1976). They analyzed the balance of buoyant force and turbulent force on a single bubble (Fig. 3-8), rather than on the whole gas region. F_B and F_T are now computed as follows:

$$F_B = g\cos\theta(\rho_L - \rho_G)\frac{\pi d_b^3}{6}, \tag{3-4}$$

$$F_T = \frac{1}{2}\rho_L \overline{v'^2}\frac{\pi \cdot d_b^3}{4} = \frac{1}{2}\rho_L (V_L^2 \frac{f_{wL}}{2})\frac{\pi d_b^3}{4}, \tag{3-5}$$

where d_b is the bubble diameter. For liquids of low viscosity, the following expression is used to approximate the stable bubble diameter d_b (Kokal & Stanislav, 1987):

$$\frac{\pi d_b^3}{6} = 1.378\left(\frac{\pi D^2}{4}V_{SG}\right)^{1.2} g^{-3/5}. \tag{3-6}$$

Again, when $F_T \geq F_B$, dispersed bubble flow takes place. The final transition criteria is:

$$V_{SL} \geq 0.8\left[\frac{\rho_L - \rho_G}{\rho_L}\frac{\cos\theta}{f_{wL}}D^{0.8}V_{SG}^{0.4}\right]^{\frac{1}{2}}. \tag{3-7}$$

Note that Eq. (3-7) includes the empirical coefficient 0.8 and V_L is replaced by V_{SL}. Thus Eq. (3-7) is more convenient to use than Eq. (3-3).

- Models for vertical flow

3) Taitel *et al.* (1980):

Taitel *et al.* (1980) proposed another method to determine the transition to dispersed bubble flow for vertical upward flow. The physical mechanism is still based on an assessment of the strength of turbulent forces. There is a critical bubble size (d_{crit}) above which the turbulent breakup process cannot prevent the bubbles from agglomerating. It is given as:

$$d_{crit} = \left(\frac{0.4\sigma}{(\rho_L - \rho_G)g}\right)^{1/2}. \tag{3-8}$$

The maximum stable diameter (d_{max}) of the dispersed phase can be obtained through the consideration of the balance between surface tension forces and turbulent forces:

$$d_{\max} = k\left(\frac{\sigma}{\rho_L}\right)^{3/5}(\varepsilon)^{-2/5} = 1.14\left(\frac{\sigma}{\rho_L}\right)^{3/5}\left(\frac{2f}{D}V_M^3\right)^{-2/5}, \tag{3-9}$$

where k is taken to be equal to 1.14, which is confirmed by experimental measurements. ε is the dissipation rate of turbulent kinetic energy. It is approximated by the friction factor f and the mixture velocity V_M. The friction factor can be calculated using the standard formula:

$$f = c\left(\frac{V_M D}{\nu_L}\right)^{-n} = 0.046\left(\frac{V_M D}{\nu_L}\right)^{-0.2}. \tag{3-10}$$

If $d_{max} > d_{crit}$, which means that the stable bubble size is too large to be broken up via liquid turbulent fluctuations, dispersed bubble flow turns to slug flow. Thus, dispersed bubble flow occurs when $d_{max} > d_{crit}$.

This criterion is further constrained by the maximum allowable volume fraction of the bubbles. For spherical bubbles arranged in a cubic lattice, the gas fraction can be at most 0.52. Therefore, regardless of how much turbulent energy is available, dispersed bubble flow cannot exist when $\alpha_G \geq 0.52$. In the region of high flow rate, the slip between the two phases can be neglected, so α_G can be replaced by the input gas volume fraction (C_G). The final transition criterion for dispersed bubble flow becomes:

$$d_{max} > d_{crit} \text{ and } C_G < 0.52, \tag{3-11}$$

where C_G is defined in Eq. (2-4).

- Models for the entire range of pipe inclinations

4) Barnea (1986):

The above transition model was further extended by Barnea (1986). Her modification includes the computations for both d_{max} and d_{crit}. The resulting model is applicable for the entire range of inclination angles.

For the critical bubble size, there are two mechanisms taken into account: the combined process of deformation and agglomeration, and migration of bubbles due to buoyancy. The first one has been addressed in Eq. (3-8), and here we call it d_{crit_D}. The second one actually is similar to the analysis in horizontal flow by Taitel & Dukler (1976), but the buoyancy is revised to act on a single bubble, as in the model by Kokal & Stanislav (1987). Consider $F_T = F_B$, where

$$F_B = g(\cos\theta)(\rho_L - \rho_G)\frac{\pi d_b^3}{6}, \tag{3-12}$$

$$F_T = \frac{1}{2}\rho_L \overline{v'^2}\frac{\pi d_b^3}{4} = \frac{1}{2}\rho_L (V_M^2 \frac{f_M}{2})\frac{\pi d_b^3}{4}. \tag{3-13}$$

The critical bubble size due to buoyancy is then obtained as:

$$d_{crit_B} = \frac{3}{8}\frac{\rho_L}{(\rho_L - \rho_G)}\frac{f_M V_M^2}{g\cos\theta}. \tag{3-14}$$

So the critical bubble size above which dispersed bubble flow cannot exist, due to either bubble coalescence or bubble migration to the top of the pipe, is given as:

$$d_{crit} = \min(d_{crit_D}, d_{crit_B}), \tag{3-15}$$

where the minimum value between d_{crit_D} and d_{crit_B} is taken.

For the maximum stable diameter of dispersed bubbles, Barnea included the effect of gas holdup on the resulting bubble size, so Eq. (3-9) was modified to the following:

$$d_{\max} = \left(0.725 + 4.15 C_G^{1/2}\right)\left(\frac{\sigma}{\rho_L}\right)^{3/5}\left(\frac{2f}{D}V_M^3\right)^{-2/5}. \tag{3-16}$$

Again, the criterion of maximum packing density also needs to be considered. The final transition criterion is the same as Eq. (3-11), except that d_{crit} is given by Eq. (3-15). The extension of d_{crit} to account for the effects of both deformation and buoyancy gives this model an advantage over the previous models in that it is applicable over the entire range of inclinations.

5) Petalas & Aziz (1998):

In the Petalas & Aziz (1998) mechanistic model, the transition to dispersed bubble flow is determined by considering the maximum packing density of liquid slugs in slug flow (slug flow consists of Taylor bubbles and liquid slugs with entrained bubbles). If the liquid fraction in the liquid slug ($\alpha_{L,s}$) is less than 0.48, transition from slug flow to dispersed bubble flow takes place. The liquid holdup within a slug body in slug flow is calculated as (Gregory *et al.*, 1978):

$$\alpha_{L,s} = \frac{1}{1+\left(\frac{V_M}{8.66}\right)^{1.39}}. \tag{3-17}$$

Again, this criterion needs to be combined wiht the maximum allowable gas volume fraction. So, dispersed bubble flow can exist when

$$\alpha_{L,s} < 0.48 \text{ and } C_G < 0.52. \tag{3-18}$$

6) Chen *et al.* (1997):

Chen et al. (1997) developed a general model for this transition. The turbulent forces in the liquid phase overcoming the gas-liquid interfacial tension is still considered to contribute to the formation of dispersed bubbles. By comparing the turbulent kinetic energy of the liquid and the surface free energy of the discrete bubbles, the transition criterion can be formulated.

The total turbulent kinetic energy of the liquid is given by:

$$E_T = \frac{3}{2}\rho_L \overline{v'^2} A V_{SL} = \frac{3}{2}\rho_L (V_{SL}^2 \frac{f_{SL}}{2}) A V_{SL}, \tag{3-19}$$

where f_{SL} is the friction factor at the superficial liquid velocity.

The total surface free energy of the dispersed gas bubbles is expressed as:

$$E_S = \frac{6\sigma}{d} Q_G = \frac{6\sigma}{d} A V_{SG}, \tag{3-20}$$

where d is the diameter of a dispersed bubble, computed as:

$$d = 2\left(\frac{0.4\sigma}{(\rho_L - \rho_G)g}\right)^{1/2}. \tag{3-21}$$

Transition to dispersed bubble flow occurs when $E_T > E_S$. The advantage of this model is that it does not require the correction for the maximum packing density at high gas flow rate.

- Model evaluation:

Comparisons between the six models discussed above and experimental data of Shoham (1982) are shown in Figs. 3-9 to 3-11. Fig. 3-9 shows the results for horizontal flow. Intermittent flow includes elongated bubble and slug flows. Here, we just show the transition boundary between dispersed bubble flow and intermittent flow. The model by Kokal & Stanislav (1987) performs better than that by Taitel & Dukler (1976), since the force balance is analyzed directly on the gas bubble. Among the three general models, Barnea's model gives the best result. Note that in the high gas flow rate, the Petalas & Aziz model and the Barnea model give the same prediction, because they both apply the maximum packing density theory. The model by Chen *et al.* (1997) does not give satisfactory predictions at low gas flow rate.

Fig. 3-10 displays the model performance for upward vertical flow. It is seen that the transition to dispersed bubble flow is not affected very much by flow orientation. The model for vertical flow by Taitel *et al.* (1980) provides comparable predictions to Barnea's model. This is understandable since the bubble deformation mechanism in the model by Taitel *et al.* is incorporated in Barnea's model. The transition boundary given by the Petalas and Aziz model deviates significantly to high liquid flow rate.

The results for downward vertical flow are given in Fig. 3-11. The experimental data indicates that the transition to dispersed bubble flow occurs at relatively low liquid flow rate. In this case, the model by Chen *et al.* (1997) displays the most accurate results.

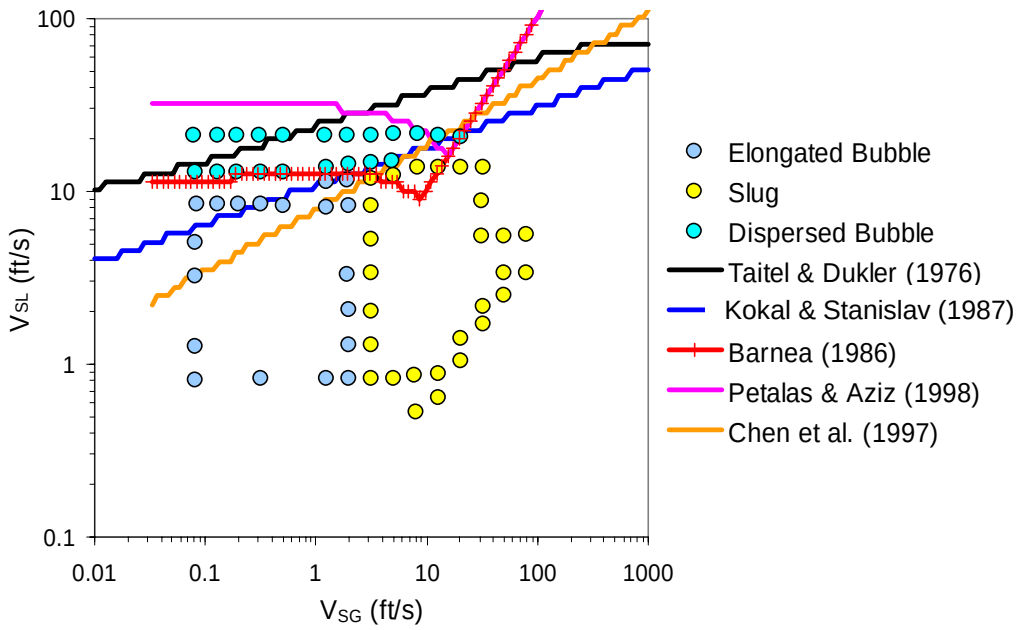


Figure 3-9: Comparison of transition models for dispersed bubble flow in horizontal flow (Data: Shoham (1982), air-water system, θ=0°, D=2.0 inch)

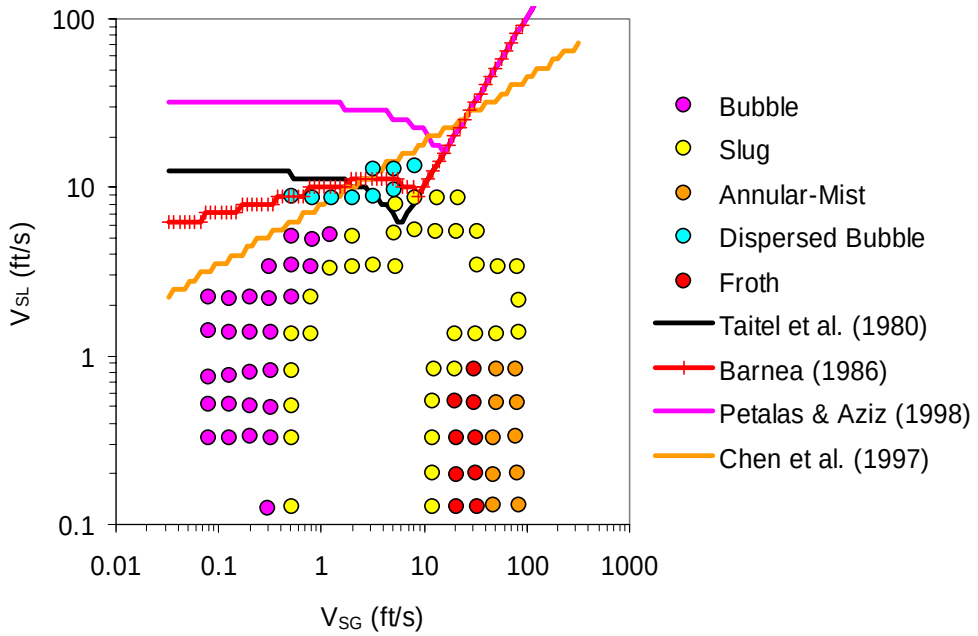


Figure 3-10: Comparison of transition models for dispersed bubble flow in vertical upward flow (Data: Shoham (1982), air-water system, θ=90°, D=2.0 inch)

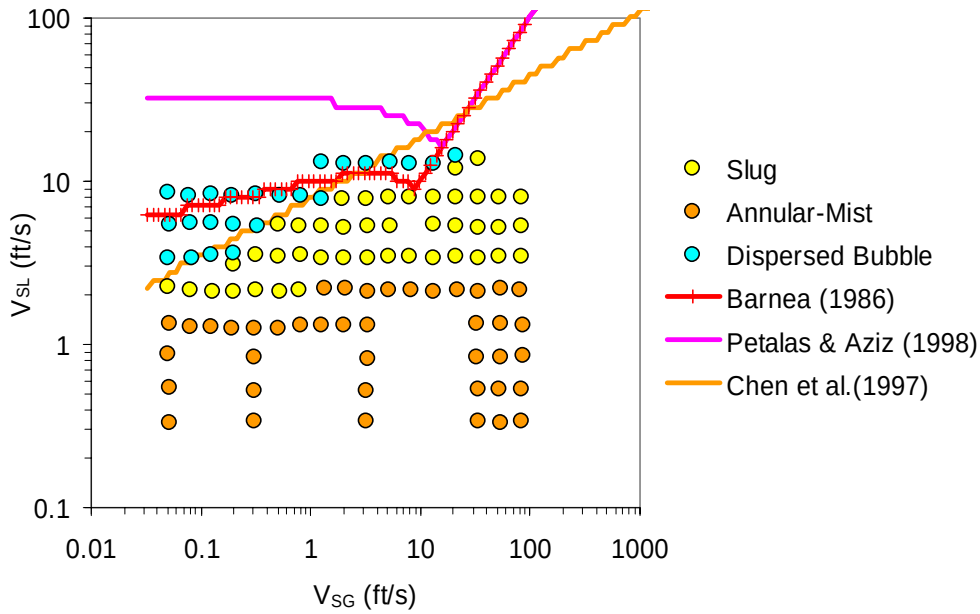


Figure 3-11: Comparison of transition models for dispersed bubble flow in vertical downward flow (Data: Shoham (1982), air-water system, θ= - 90°, D=2.0 inch)

Most of the models discussed in this section were developed from the consideration of turbulent fluctuation forces breaking up the gas phase into discrete bubbles. Overall, the Barnea (1986) model predicts the most accurate transition boundary. This is not surprising since her model accounts for more physical mechanisms than the other models. The Petalas & Aziz (1998) model is based on the liquid volume fraction in the slug body in intermittent flow. It is possible, however, that the performance of this model could be improved by tuning the critical value used in the model.

3.3.2. Interfacial Friction Factor in Stratified Flow

We next investigate the transition between stratified flow and intermittent flow. The prediction to this transition by the Petalas & Aziz (1998) mechanistic model also shows some inaccuracy (See Figs. 3-1 to 3-4). This transition is based on a Kelvin-Helmholtz wave stability analysis, which is finally determined by the value of the liquid height h_L (Fig. 3-12). The determination of h_L comes from the solution of the coupled momentum equations for the liquid and gas phases:

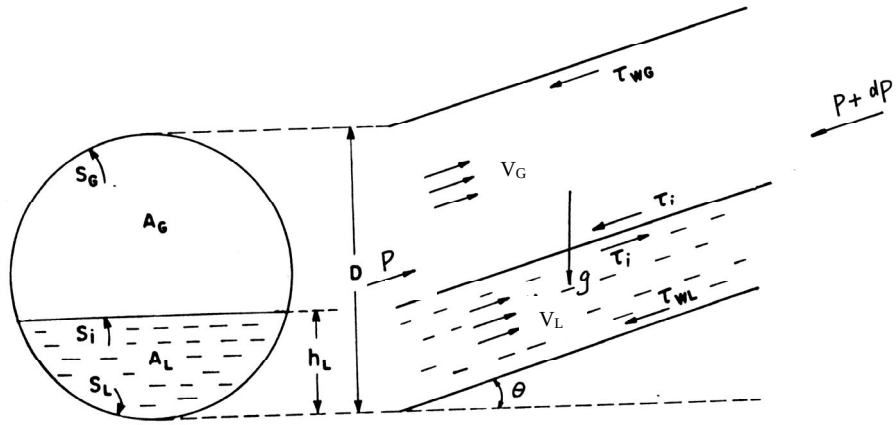


Figure 3-12: Schematic of stratified flow (modified from Shoham, 1982)

$$\begin{aligned} -A_L\left(\frac{dp}{dx}\right) - \tau_{wL}S_L + \tau_i S_i - \rho_L A_L g\sin\theta = 0 \\ -A_G\left(\frac{dp}{dx}\right) - \tau_{wG}S_G - \tau_i S_i - \rho_G A_G g\sin\theta = 0 \end{aligned}, \tag{3-22}$$

where A_L, A_G, S_L, S_G, and S_i are geometric parameters that only depend on h_L. The quantities τ_{wL}, τ_{wG}, and τ_i are shear stresses. They are calculated by empirical correlations for the friction factors. Among them, τ_i is the shear stress between the liquid phase and the gas phase. It is associated with the interfacial friction factor f_i. Finally, in Eq. (3-22), there are two unknowns dp/dx and h_L. They can be determined by the two equations. Therefore, h_L depends on f_i, as does the transition from stratified to intermittent flow.

The simplest expression for f_i is to approximate it using the wall friction factor for the gas phase. This may be reasonable since the gas-liquid interface can be thought of as a smooth surface.

Petalas & Aziz (1998) developed the following correlation using the data in the Stanford Multiphase Flow Database:

$$f_i = \left(0.004 + 0.5\times10^{-6}\,\mathrm{Re}_{SL}\right)Fr_L^{1.335}\left(\frac{\rho_L D_G}{\rho_G V_G^2}\right), \tag{3-23}$$

where D_G is the gas hydraulic diameter, $\mathrm{Re}_{SL} = \dfrac{D\rho_L V_{SL}}{\mu_L}$ and $Fr_L = \dfrac{V_L}{\sqrt{gh_L}}$.

We apply these two correlations in the Petalas & Aziz (1998) mechanistic model and get the transition boundaries as shown in Fig. 3-13. We see that hump in the original transition curve is due to f_i. It can be eliminated by using the simple approximation of $f_i = f_{wG}$.

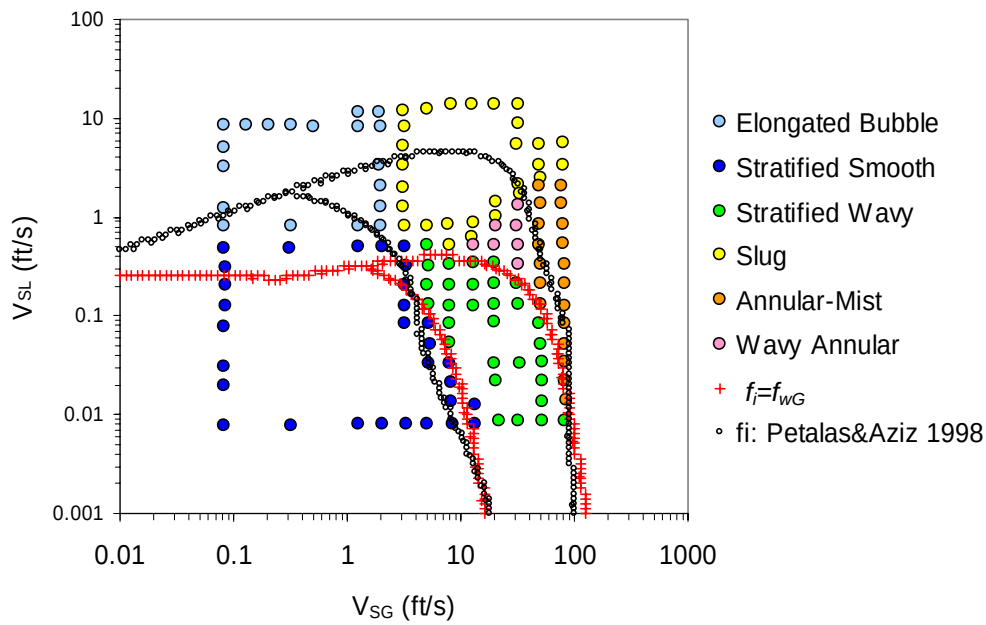


Figure 3-13: Effects of interfacial friction factor for horizontal flow (Data: Shoham (1982), air-water system, θ= 0°, D=2.0 inch)

In the literature, there are numerous expressions for f_i. Ouyang (1995) reviewed 26 published correlations for f_i, and developed a new one based on the data in the SMFD. We now assess the use of Ouyang's correlation (1995) and that by Baker *et al.* (1988) in the Petalas & Aziz (1998) mechanistic model. The Baker *et al.* (1988) correlation is rather complicated; a detailed description can be found in Ouyang (1995). The Ouyang (1995) correlation is given as:

$$f_i = 10^{-8.0942+4.2893\alpha_L \sin\theta} \frac{f_{wL}^{0.8732} N_{vL}^{0.3072} N_D^{1.0365}}{N_{\mu G}^{1.9140} H^{0.9783}}, \tag{3-24}$$

where N_{vL} is the liquid velocity number, $N_{\mu G}$ the gas viscosity number, N_D the pipe diameter number and H the holdup ratio. For more details, see Ouyang (1995).

The comparison of the four correlations for f_i described above is illustrated in Fig. 3-14. The f_i we discuss here is for stratified flow, so it only has influence on the transition from stratified flow. We see that the Ouyang (1995) correlation and that of Petalas & Aziz (1998) give similar predictions. They both tend to deviate to high liquid flow rate. On the other hand, the Baker *et al.* (1988) model and the simplest approximation of $f_i=f_{wG}$ perform about the same, displaying more accurate predictions than the models by Ouyang (1995) and Petalas & Aziz (1998).

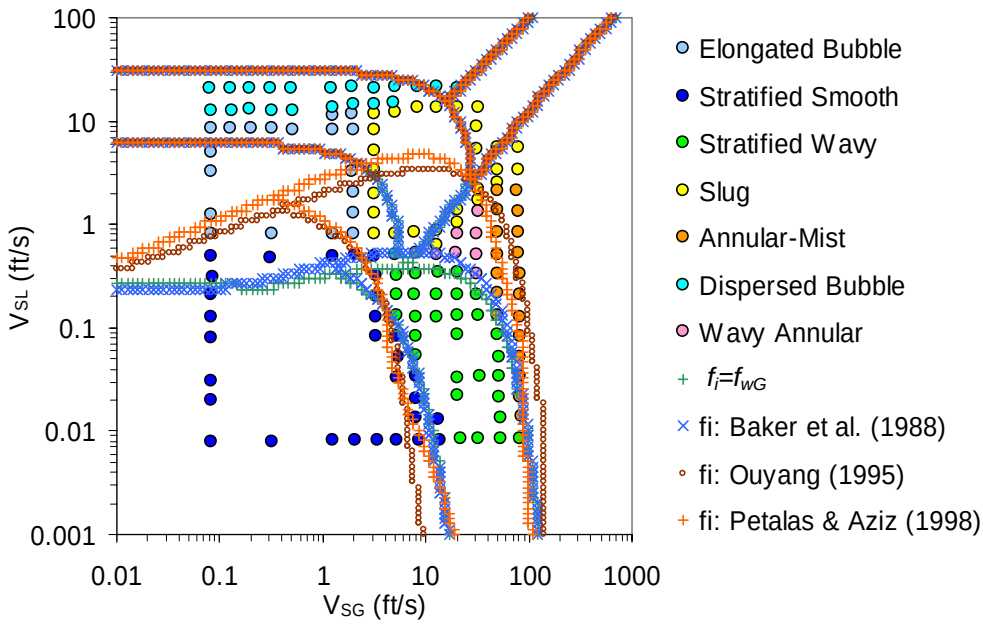


Figure 3-14: Comparison of models with different f_i (Data: Shoham (1982), air-water system, θ= 0°, D=2.0 inch)

Stratified flow often occurs in downward flow, so it is important to be able to predict accurately this transition boundary in this case. Comparison between the model and data for -10° downward flow is presented in Fig. 3-15. Due to the expansion of the stratified flow pattern, the intermittent flow region shrinks. The two correlations provide comparable results, with the model of $f_i=f_{wG}$ displaying a flat trend at low gas flow rate, which is also illustrated by the data. The expansion of the stratified flow region occurs primarily in downward flow from 0° to -10°. From -10° to -70°, this region is almost unchanged. Increasing the downward angle from -10°, the annular region expands and the

stratified flow region shrinks until it disappears completely at vertical downward flow (Shoham, 1982). The transition in vertical downward flow is illustrated in Fig. 3-16, in which only three flow regimes are observed experimentally. Using the Petalas & Aziz (1998) correlation for f_i, a considerable stratified region is predicted, which is not consistent with experimental observations. The correlation of $f_i=f_{wG}$ provides a more reasonable result.

In summary, the interfacial friction factor f_i in stratified flow plays an important role in the transition between intermittent flow and stratified flow, as well as on the calculations of pressure drop and holdup. Based on Shoham's data (1982), the simple correlation $f_i=f_{wG}$ gives better results in terms of stratified flow prediction, though it has been shown previously (Ouyang, 1995, Petalas & Aziz, 1998) that the correlations by Ouyang (1995) and Petalas & Aziz (1998) give better results for the calculations of pressure drop and holdup. Therefore, $f_i=f_{wG}$ is recommended for the purpose of flow pattern prediction, but not for the calculation of pressure drop and holdup.

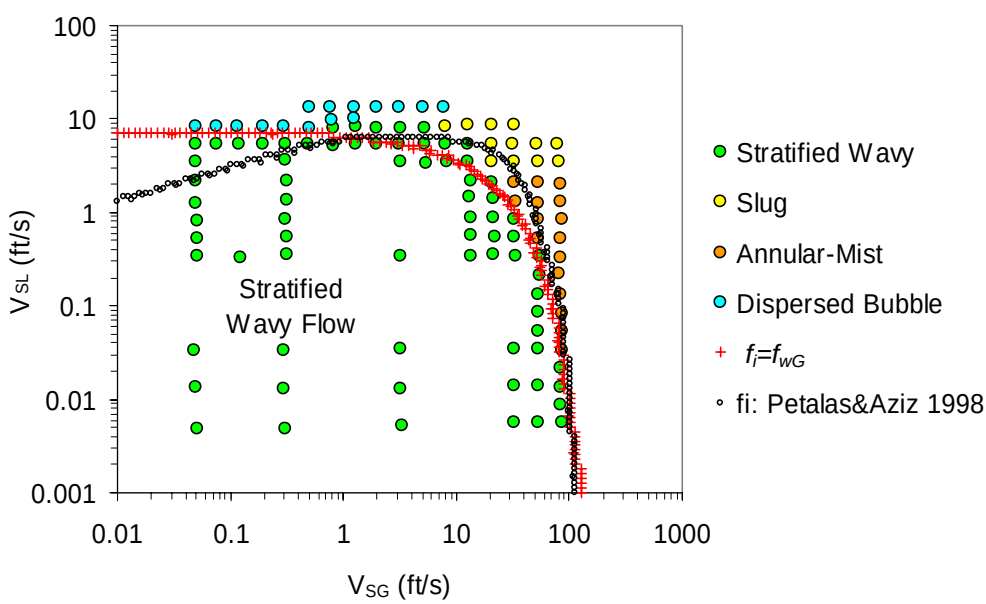


Figure 3-15: Effects of interfacial friction factor for downward flow (Data: Shoham (1982), air-water system, θ= -10°, D=2.0 inch)

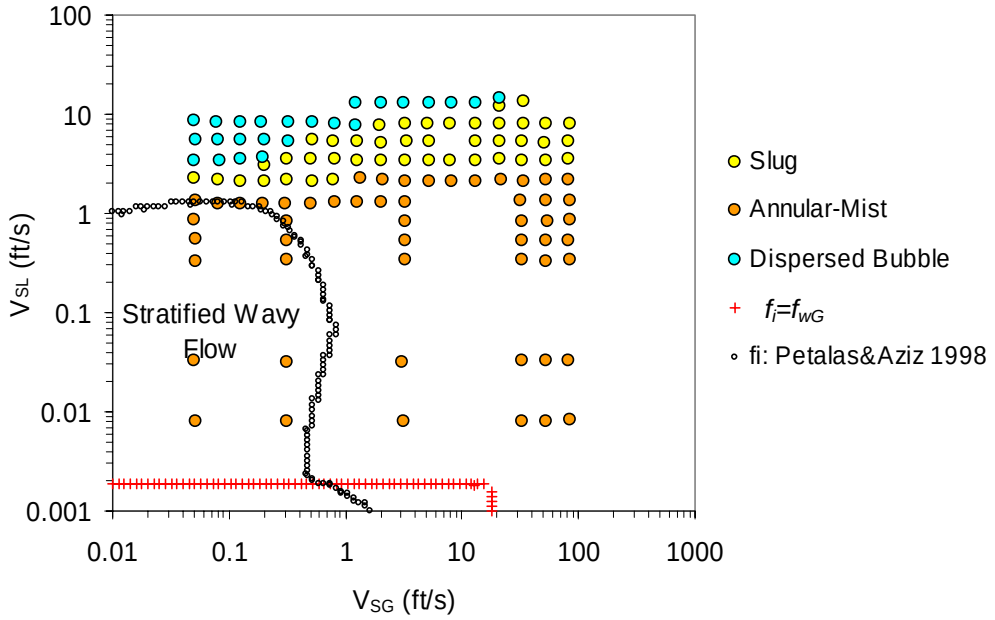


Figure 3-16: Effects of interfacial friction factor for vertical downward flow (Data: Shoham (1982), air-water system, θ= -90°, D=2.0 inch)

3.3.3. Transition to Annular-Mist Flow

As shown in Fig. 3-1, there are three main flow pattern transitions. The transition to dispersed bubble flow and the transition between intermittent and stratified flows have been discussed. We now consider the transition to annular-mist flow. This flow pattern can be obtained either from stratified flow or from intermittent flow.

Our discussion of the transition between stratified flow and intermittent flow can be extended to predict the transition between stratified flow and annular flow. For horizontal flow, Taitel & Dukler (1976) proposed that the transition of stratified flow to slug flow or annular flow depends on the liquid holdup. If the liquid level is low, annular flow results, while if the liquid level is high enough to form a complete bridge across the pipe, slug flow will occur. This analysis results in the transition boundary to stratified flow shown in Fig. 3-17 (it is obtained using the Petalas & Aziz (1998) mechanistic model with interfacial friction factor f_i=f_{wG} in stratified flow).

In this section, we will focus on the latter case --- transition to annular-mist flow from intermittent flow. The fast moving gas core preventing the liquid film from falling down is the essential feature of vertical annular-mist flow.

Taitel *et al.* (1980) considered the balance between the gravity and drag forces acting on a liquid droplet in the gas core and obtained the following criterion:

$$V_{SG} \geq \frac{3.1\left[\sigma g\left(\rho_L - \rho_G\right)\right]^{0.25}}{\rho_G^{\ 0.5}} . \tag{3-25}$$

The physical mechanism here is that the annular-mist flow cannot exist unless the gas velocity in the gas core is sufficient to lift the entrained liquid droplets.

Based on the same mechanism, McQuillan & Whalley (1985) considered the Froude number, which is the ratio of inertia force to gravity force to get the following expression:

$$V_{SG} \geq \frac{\left[gD\left(\rho_L - \rho_G\right)\right]^{0.5}}{\rho_G^{\ 0.5}} . \tag{3-26}$$

For a given system, Eqs. (3-25) and (3-26) result in transitions at constant superficial gas velocities; i.e., straight lines on the flow pattern map.

Joseph *et al.* (1996) proposed another criterion for this transition for horizontal and vertical flows. They interpreted the high gas flow rate in annular-mist flow in terms of effective viscosity --- the stable annular flow appears only when the gas core is very highly turbulent with a higher effective viscosity than the liquid in the annulus. Their transition criterion is given by:

$$\frac{\rho_G V_{SG} D}{\mu_L} = \begin{cases} 1000 & if \quad \dfrac{\rho_L V_{SL,\delta} D}{\mu_L} \leq 2000 \\ \dfrac{1}{2}\dfrac{\rho_L V_{SL,\delta} D}{\mu_L} & if \quad \dfrac{\rho_L V_{SL,\delta} D}{\mu_L} > 2000 \end{cases} , \tag{3-27}$$

where $V_{SL,\delta}$, is the superficial velocity of the liquid film, approximated by $V_{SL,\delta} = 0.05 V_{SL}$ in this case.

A general model was presented by Barnea (1986), which is based on two conditions. The first mechanism is the instability of the liquid film. The minimum interfacial shear stress is associated with a change in the direction of the velocity profile in the film. This is only valid for vertical upward flow. Another mechanism is the spontaneous blockage of the gas core due to a large supply of liquid from the film. The transition from annular-mist flow to slug flow will take place when the liquid holdup exceeds one half of the value associated with the maximum volumetric packing density (0.52), that is:

$$\alpha_L \geq \frac{1}{2}(1-0.52) = 0.24\,. \tag{3-28}$$

This transition model (combination of the two mechanisms) is widely used in a variety of mechanistic models. In the work of Ansai *et al.* (1994), Barnea's model was modified using correlations accounting for the liquid entrainment in the gas core. Similarly, in the Petalas and Aziz (1998) mechanistic model, the effects of both liquid entrainment and pipe roughness are included.

Kaya *et al.* (1999) considered the transition from intermittent flow to annular flow to occur at a critical void fraction: $\alpha_G > 0.75$. Note that this is essentially the same as Eq. (3-28). Kaya *et al.* essentially simplified the transition criterion by only considering the second mechanisms in Barnea's model. However, to obtain the in situ α_G, momentum equations for annular-mist flow, similar to those for stratified flow presented in Eq. (3-19) need to be solved. In the Kaya *et al.* (1999) model, the standard expression for the friction factor is used and the entrained liquid droplets in the gas core are neglected.

Next we will evaluate the above models; specifically the Joseph *et al.* (1996) model, the Petalas & Aziz (1998) model and the Kaya *et al.* (1999) model. In addition, we will calculate α_G in the Petalas & Aziz (1998) mechanistic model and apply the critical void fraction 0.75 to this calculated α_G. This is expected to be different from that of Kaya *et al.* (1999), since the correlations in each model are different. The results will allow us to assess the relative importance of the two mechanisms Barnea (1986) has proposed, as well as to evaluate the effects of correlations for annular-mist flow.

The results for horizontal flow are shown in Fig. 3-17. The transition to stratified flow is also shown in order to illustrate that annular-mist flow can also transition from stratified flow. In the procedure of determining the flow patterns, prediction to stratified flow is performed before that to annular-mist flow. With respect to the transition to annular-mist flow, all models present reasonable prediction. The Petalas & Aziz (1998) model (which incorporates the mechanisms by Barnea, 1986) and the use of the critical void fraction gives the same results, which means that the spontaneous blockage mechanism in Barnea's model is applied here rather than the mechanism involving instability of the liquid film. The difference of this prediction from that by the Kaya *et al.* (1999) model results from the different correlations used. The same observations can be made in Fig. 3-18, in which the transition to annular-mist flow in an upward vertical orientation is shown. The effective viscosity criterion by Joseph *et al.* predicts the transition boundary most accurately.

However, this is not the case for vertical downward flow as shown in Fig. 3-19. In -90° downward flow, annular-mist flow takes place in most flow conditions. Joseph's model is only applicable for horizontal and upward flow. Though the results from the Petalas & Aziz (1998) model and Kaya's model are also not entirely satisfactory, they provide a reasonable trend. We believe that changes in correlations used in these models, resulting in changes in the in situ gas volume fraction, should affect this transition.

The evaluation of all of these models shows that none of the existing transition models gives an entirely satisfactory prediction over the entire range of pipe inclinations. The holdup based transition criterion is reasonable and is recommended for this transition. However, like the effect of the interfacial friction factor in stratified flow, correlations such as the interfacial friction factor in annular-mist flow and the liquid volume fraction in the gas core may have a strong impact on this transition boundary.

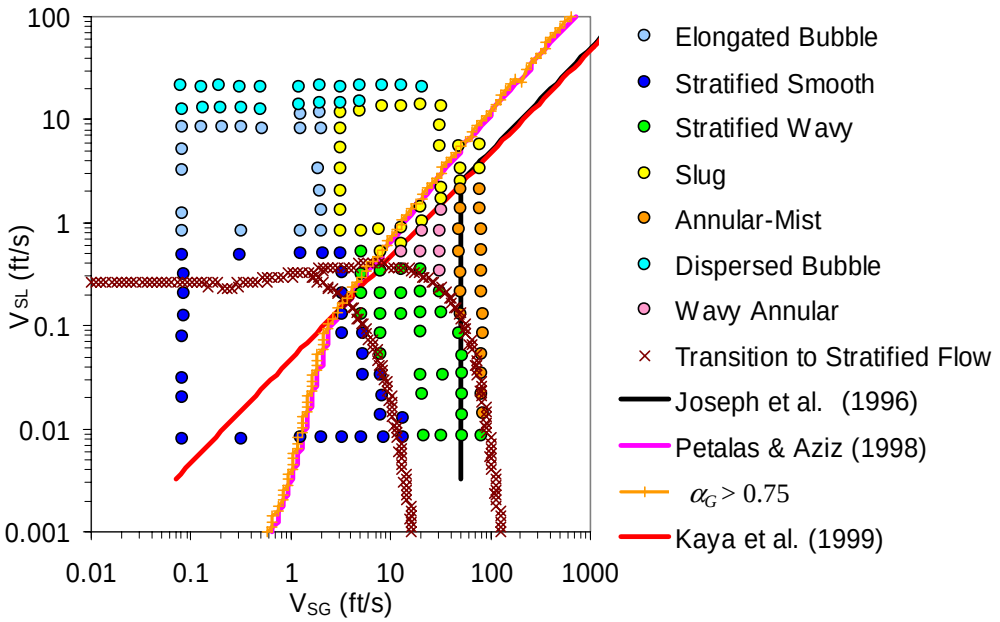


Figure 3-17: Comparison of transition models for annular-mist flow (Data: Shoham (1982), air-water system, θ= 0°, D=2.0 inch)

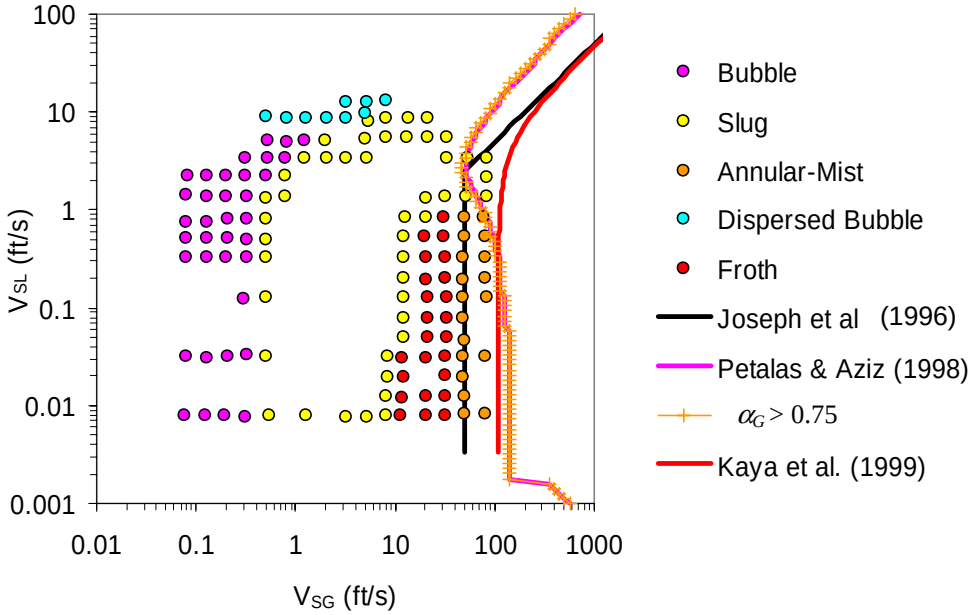


Figure 3-18: Comparison of transition models for annular-mist flow (Data: Shoham (1982), air-water system, θ= 90°, D=2.0 inch)

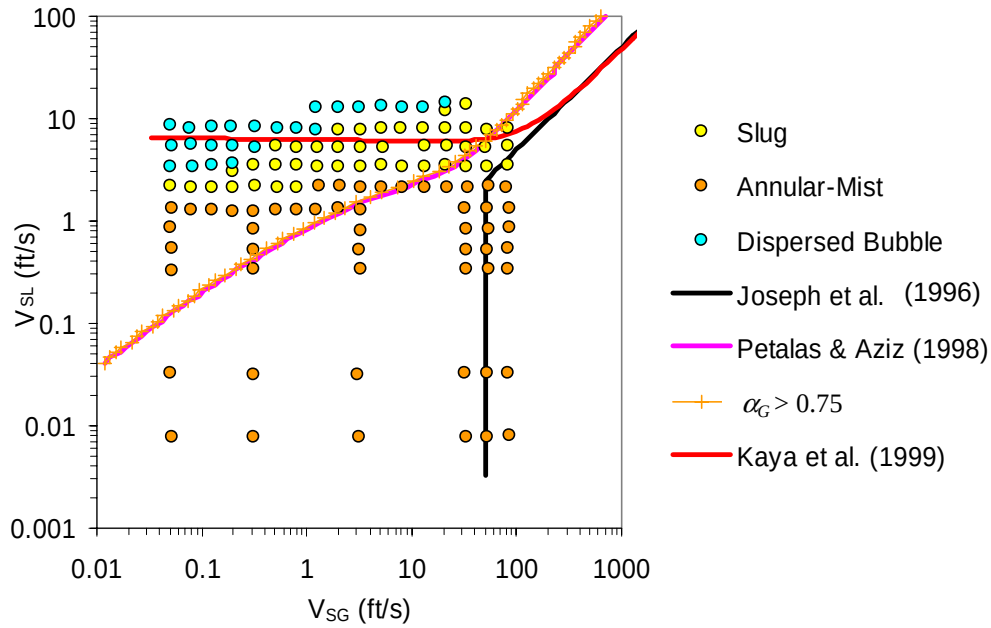


Figure 3-19: Comparison of transition models for annular-mist flow (Data: Shoham (1982), air-water system, θ= -90°, D=2.0 inch)

In this chapter, we have discussed the flow pattern transition predictions in mechanistic models, specifically, the Petalas & Aziz (1998) mechanistic model. Based on the data considered, overall this model provides reasonably good predictions. We also investigated other transition criteria and gave recommendations to improve the current mechanistic model. This includes Barnea's (1986) model for transition to dispersed bubble flow and the use of f_{wG} to approximate f_i in stratified flow. The Petalas & Aziz (1998) holdup based transition criterion to dispersed flow gives the correct trend, but the critical value needs to be adjusted or other correlations for liquid holdup in the slug used. The holdup transition is more accurate in the prediction to annular-mist flow. However, the correlations used in the mechanistic model have a strong effect on the results. The effects of fluid properties were found to be less significant, in their effects on transitions, than the pipe inclination.

Chapter 4

4. Investigation of Drift-Flux Model Parameters

In this chapter, the drift-flux model (DFM) is applied to different flow patterns. Based on our observations, we model the parameters C_0 and V_d in DFM as linear functions of the gas volume fraction α_G. A method is proposed to determine these two parameters by matching the experimental α_G and the α_G calculated from DFM as closely as possible. The physical meanings of C_0 and V_d are illustrated by analyzing the resulting correlations. Finally, comparisons between the drift-flux model correlations in Eclipse (Schlumberger GeoQuest, 2000) and the experimental data are presented.

4.1. Drift-Flux Model Parameters

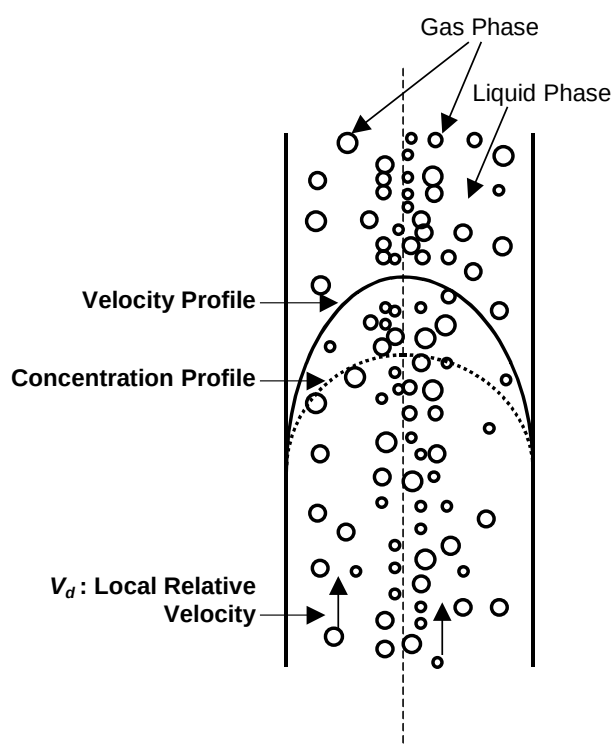


Figure 4-1: Schematic of velocity and concentration profiles

The drift-flux model proposed by Zuber & Findlay (1965) can be used to calculate the gas volume fraction and interpret holdup data. It correlates the actual gas velocity V_G and the mixture velocity V_M, using two parameters C_0 and V_d:

$$V_G = \frac{V_{SG}}{\alpha_G} = C_0 V_M + V_d \,, \tag{4-1}$$

where V_M is the mixture velocity as defined in Eq. (2-2). C_0 is referred to as the distribution parameter or profile parameter. It accounts for the effects of the non-uniform distribution of both velocity and concentration profiles (see Fig. 4-1 for typical gas concentration and velocity distributions). If the two phases are uniformly mixed, the concentration profile will be flat and C_0 should be equal to one. V_d is called the drift velocity of gas, and accounts for the local relative velocity between the two phases. If the liquid is stationary, V_d corresponds to the gas rise velocity in the stagnant liquid.

With the two parameters and superficial velocities, the in situ gas volume fraction can be calculated. The accuracy of the predicted α_G depends on the use of appropriate values for C_0 and V_d.

4.2. Drift-Flux Model in Different Flow Patterns

Traditionally, the drift-flux model is used most widely for vertical dispersed system. Eq. (4-1) is derived from the continuity equation in dispersed systems (Govier & Aziz, 1972). Nonetheless, the linear relationship between V_G and V_M has been confirmed empirically for flow regimes other than dispersed flow. These even include separated horizontal flows (Franca & Lahey, 1992).

In Figs. 4-2 and 4-3, we plot V_G vs. V_M in different flow patterns for both horizontal and vertical flows. Using linear regression, we determine the two parameters C_0 and V_d. Very high degrees of correlation for V_G and V_M are observed for all the flow conditions (values of R^2 greater than 0.98 in all cases, where R^2 is the square of correlation coefficient).

Since different flow mechanisms operate in different flow patterns, the DFM parameters C_0 and V_d should depend on the flow regimes and flow orientations. If we consider

vertical flow, we note that the value of C_0 is close to 1 in annular-mist flow, while it is about 1.16 in slug flow. This behavior can be explained through consideration of the concentration profiles in annular-mist and slug flows. In annular-mist flow, although there are a few liquid droplets entrained in the gas core, the overall gas distribution is fairly uniform. Thus, $C_0 \sim 1$. However, in slug flow where Taylor bubbles and liquid slugs appear alternatively, the non-uniform effects are much stronger, which gives rise to $C_0 > 1$.

Figs. 4-4 and 4-5 summarize the values of C_0 and V_d obtained in various flow patterns in vertical flow. We calculate the average gas volume fraction in each flow pattern and use it for the presentation of the results. The lowest gas volume fraction occurs in the elongated bubble flow. The gas volume fraction increases in slug flow as the entrained gas bubbles can exist in the liquid slug. The highest void fraction is in annular-mist flow because of the existence of the gas core. There are significant differences in the values of C_0 and V_d among the various flow patterns.

The relation between flow pattern information and volume fraction can be demonstrated more clearly via flow pattern maps. Instead of using V_{SL} vs. V_{SG} as shown in Chapter 3, we use the coordinates of V_M vs. α_G in Figs. 4-6 and 4-7 to represent the information on flow patterns. Again, different colors represent different flow patterns.* For vertical flow, at low α_G and V_M, elongated bubble flow is observed. As α_G and V_M increase, slug and churn flows occur, and finally annular-mist flow is achieved. In horizontal flow, stratified flow is also associated with high gas volume fractions (Fig. 4-6). However, it can be differentiated from the intermittent and annular-mist flows by its lower flow rate. This can also be seen in the flow pattern maps in Figs. 3-1 to 3-3. The flow regimes in vertical flow are relatively simple due to the absence of stratified flow. Therefore, α_G itself can provide us with reasonably accurate information on flow patterns (Fig. 4-7).

* The color code for each flow pattern used in this chapter is different than that used in Chapter 3. This is done to achieve a more visible representation of the different flow patterns in the scatter plot of α_G, shown later in this chapter.

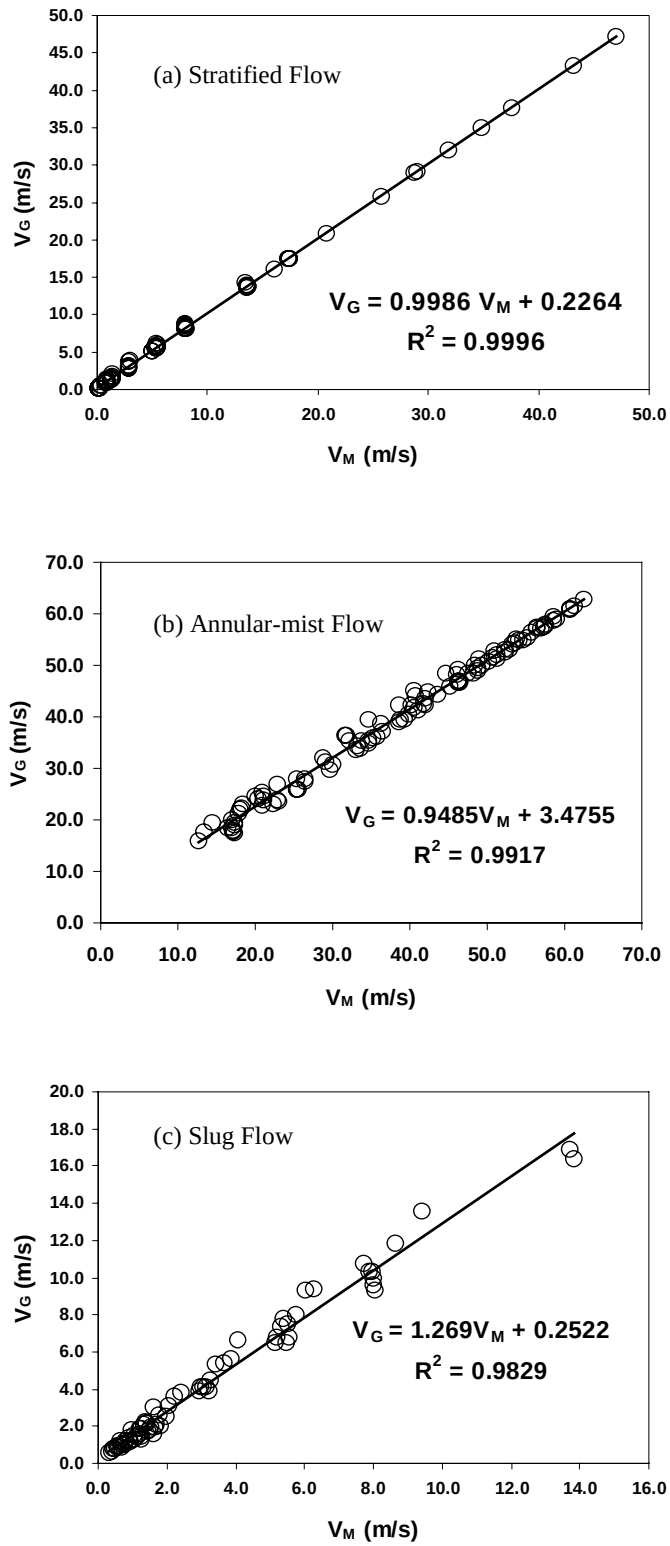


Figure 4-2: Drift-flux model in horizontal flow (Data: Spedding & Nguyen (1976), air-water system, *D*=1.79 inch)

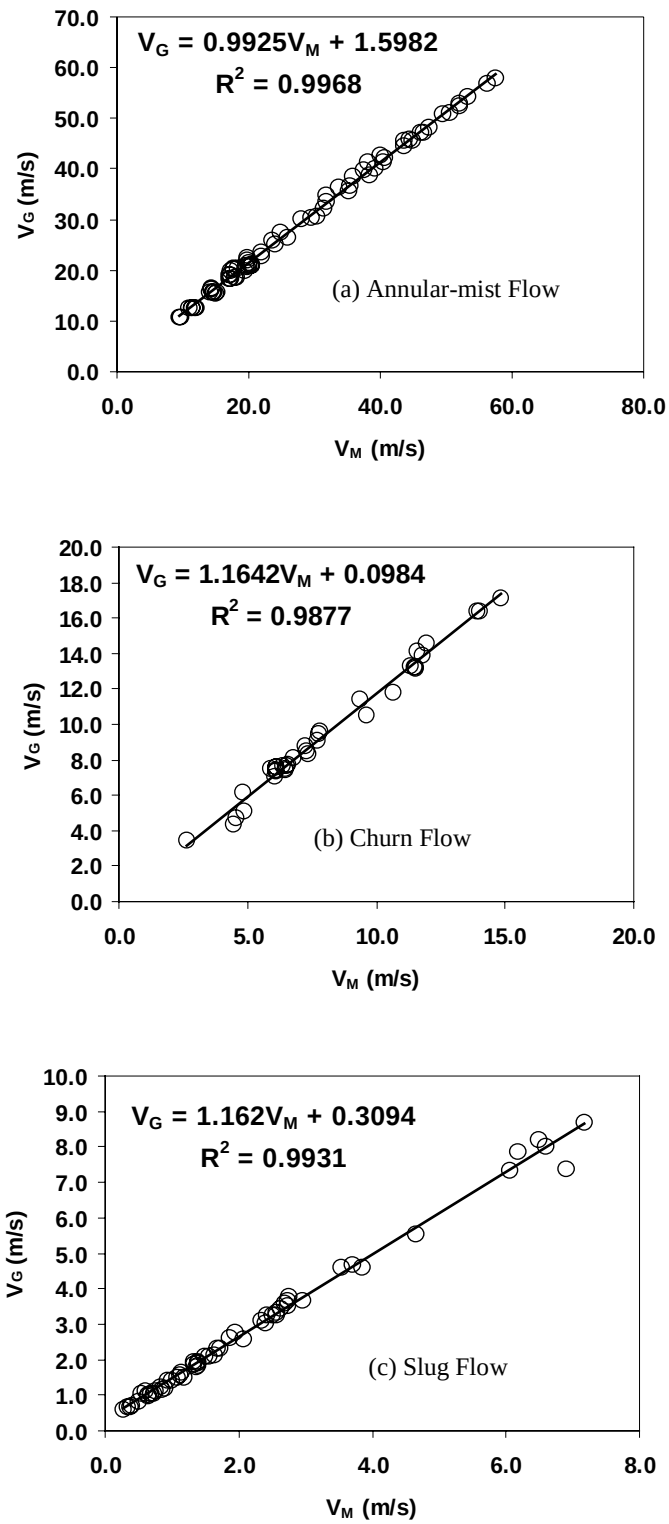


Figure 4-3: Drift-flux model in vertical flow (Data: Spedding & Nguyen (1976), air-water system, *D*=1.79 inch)

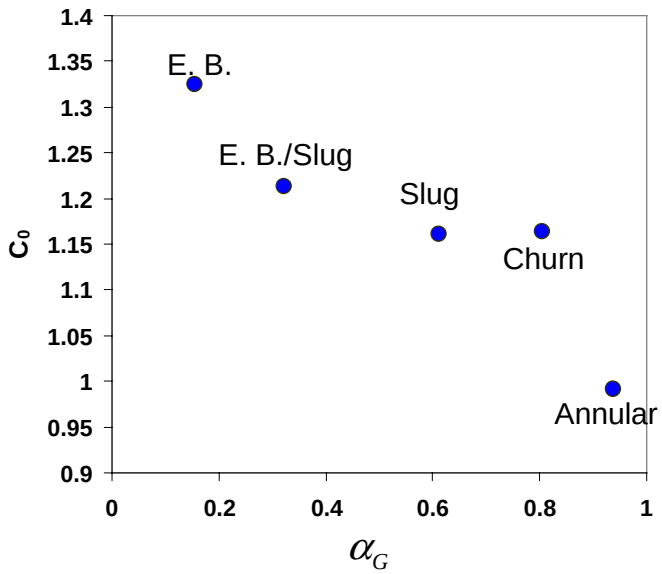


Figure 4-4: Values of C_0 in different flow patterns (Data: Spedding & Nguyen (1976), air-water system, vertical flow, D=1.79 inch)

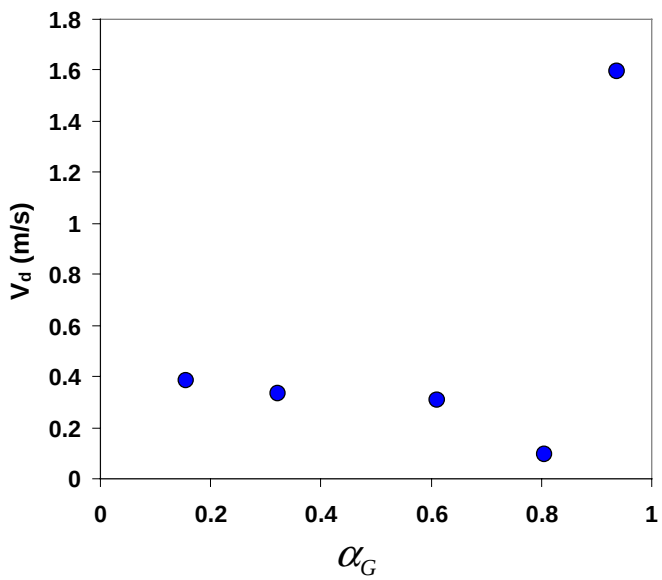


Figure 4-5: Values of V_d in different flow patterns (Data: Spedding & Nguyen (1976), air-water system, vertical flow, D=1.79 inch)

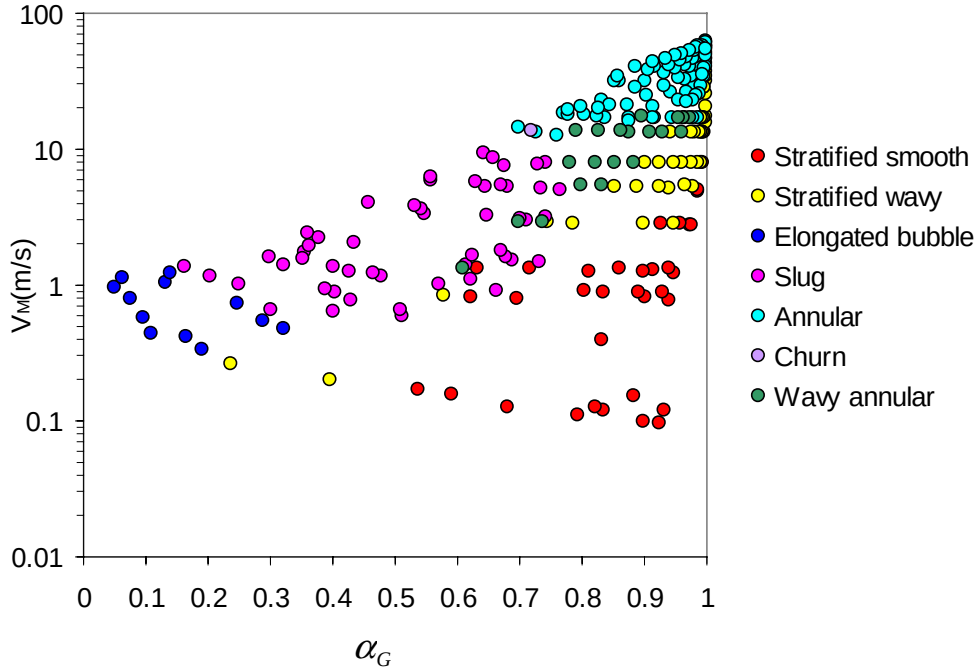


Figure 4-6: Flow pattern map in coordinates of V_M vs. α_G (Data: Spedding & Nguyen (1976), air-water system, horizontal flow, D=1.79 inch)

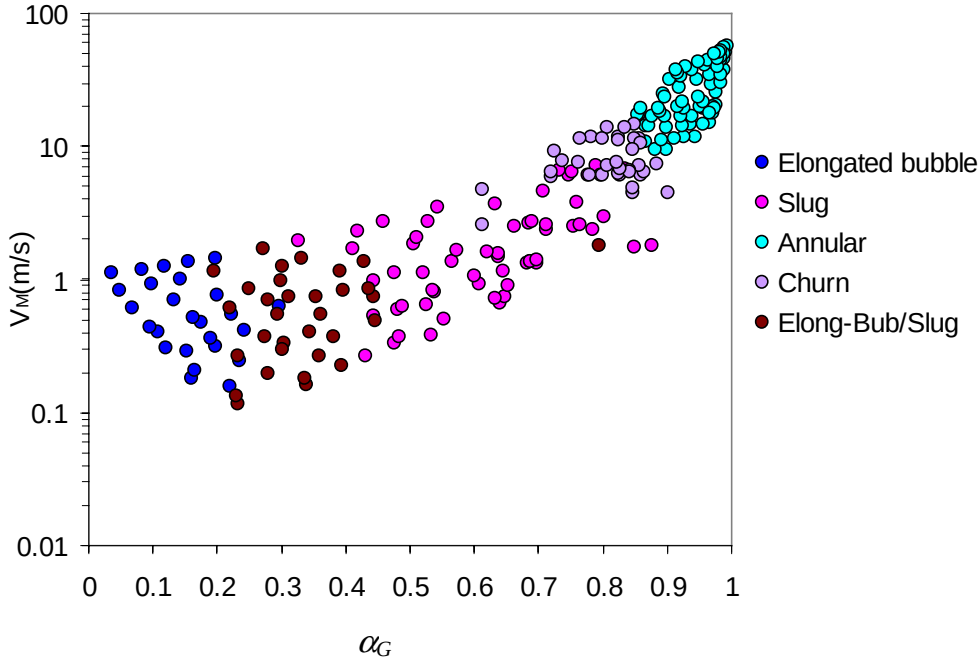


Figure 4-7: Flow pattern map in coordinates of V_M vs. α_G (Data: Spedding & Nguyen (1976), air-water system, vertical flow, D=1.79 inch)

We saw in Figs. 4-2 to 4-7 that flow patterns have a strong influence on the DFM parameters. Thus it is not adequate to take these two parameters as constants. We also saw that α_G provides some indication of the flow pattern. In next section, we will describe our estimation of C_0 and V_d based on α_G.

4.3. Method for Parameter Determination

4.3.1. Objective Function using α_G

With the DFM parameters we obtained by linear regression, we can calculate the in situ gas volume fraction:

$$\alpha_G = \frac{V_{SG}}{C_0 V_M + V_d}. \tag{4-2}$$

Fig. 4-8 (a) shows the scatter plot between the predicted α_G and the experimental α_G*. Different colors represent data points in different flow patterns, where the color code is the same as that in Fig. 4-7. Fig. 4-8(b) displays the comparison between the experimental V_G and predicted V_G. Compared to the excellent agreement of V_G in Fig. 4-8(b), the prediction for α_G is much less accurate.

The aim of the procedure applied above is to determine C_0 and V_d by minimizing the error of the following objective function:

$$E_{V_G} = \sum_{i=1}^{m} \left[\left(\frac{V_{SG}}{\alpha_G} \right)_i - \left(C_0 V_{Mi} + V_d \right) \right]^2, \tag{4-3}$$

where m is the total number of experimental data points. Since our objective here is to predict the in situ gas volume fraction as accurately as possible, a more appropriate way of determining C_0 and V_d is to directly minimize the error between the measured α_G and the estimated α_G:

* In a scatter plot, the correlation coefficient ρ is used to quantify the linear dependence between two variables. For the two variables X and Y, ρ is defined as: $\rho=\sigma^2_{XY}/(\sigma_X\sigma_Y)$, where σ^2_{XY} is the covariance between X and Y, σ_X and σ_Y the standard deviation of X and Y respectively. The previously used R^2 in the $V_G \sim V_M$ plot is given by $R^2=\rho^2$.

$$E_{\alpha_G} = \sum_{i=1}^{m} \left[\alpha_{Gi} - \left(\frac{V_{SGi}}{C_0 V_{Mi} + V_d} \right) \right]^2 . \tag{4-4}$$

Minimization of Eq. (4-4) is a nonlinear least square problem. A Gauss Newton algorithm (Li *et al.*, 1995) is used to solve this system.

Using C_0 and V_d determined from Eq. (4-4), the comparisons between the predicted and measured values for both α_G and V_G are presented in Fig. 4-9. Significant improvement is obtained for the calculation of α_G except at very high α_G. Note that from the two methods (Eqs. (4-3) and (4-4)), we get different values for C_0 and V_d. However, in Figs. 4-8(b) and 4-9(b), we see that the accuracy of V_G for these two methods is the same, and both have very high correlation coefficients. The discrepancies between the predictions for α_G and V_G indicate that V_G is less sensitive to the values of the DFM parameters. In other words, the actual gas velocity is not a good indicator of how well the drift-flux model works for the prediction of the in situ gas volume fraction.

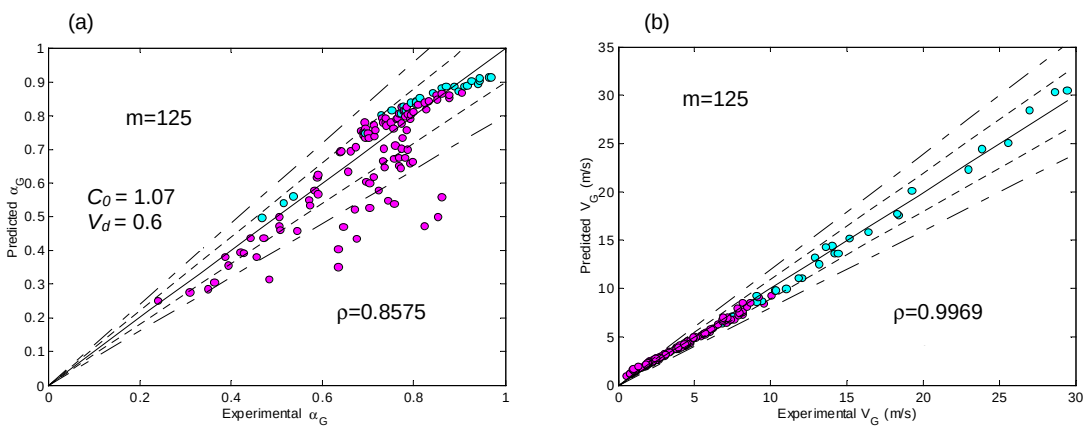


Figure 4-8: Prediction results using E_{V_G} (Data: SU66, Govier *et al.* (1957), air-water system, vertical flow, D=1.02 inch)

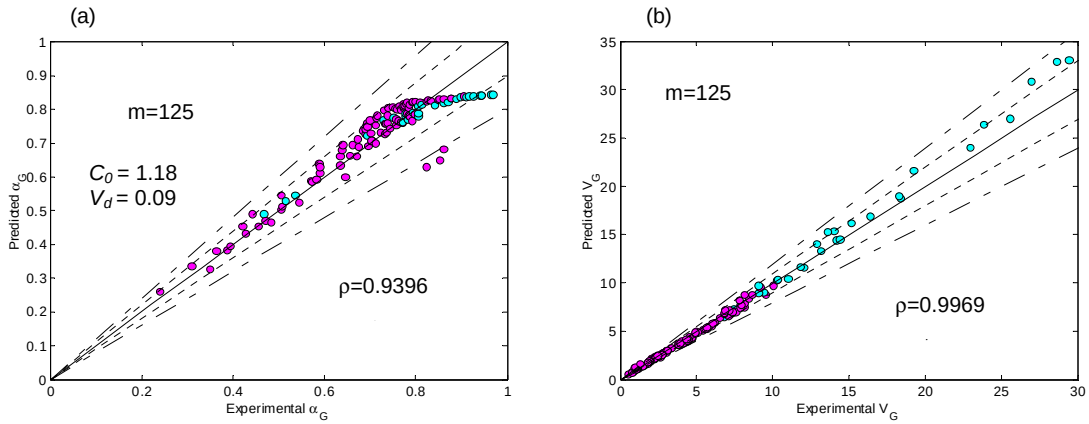


Figure 4-9: Prediction results using E_{α_G} (Data: SU66, Govier *et al.* (1957), air-water system, vertical flow, *D*=1.02 inch)

The impact of flow patterns on the values of C_0 and V_d was shown in section 4.2. Instead of calculating these two parameters using the data over the entire range of interest, we can apply Eq. (4-3) and Eq. (4-4) to each flow pattern. Fig. 4-10 displays this result using the objective function E_{V_G}, while Fig. 4-11 is for the objective function E_{α_G}. Again, a better match is obtained when E_{α_G} is used (Figs. 4-10(a) and 4-11(a)). One interesting observation is that when the flow pattern information is taken into account, E_{V_G} can give results comparable to those using E_{α_G} (Figs. 4-10(b) and 4-11(b)). However, it should be kept in mind that we generally do not know flow pattern information in advance. Thus we conclude that E_{α_G} is the better way of determining C_0 and V_d. In the next section, α_G will be introduced to indicate the flow patterns and E_{α_G} will be used as the objective function.

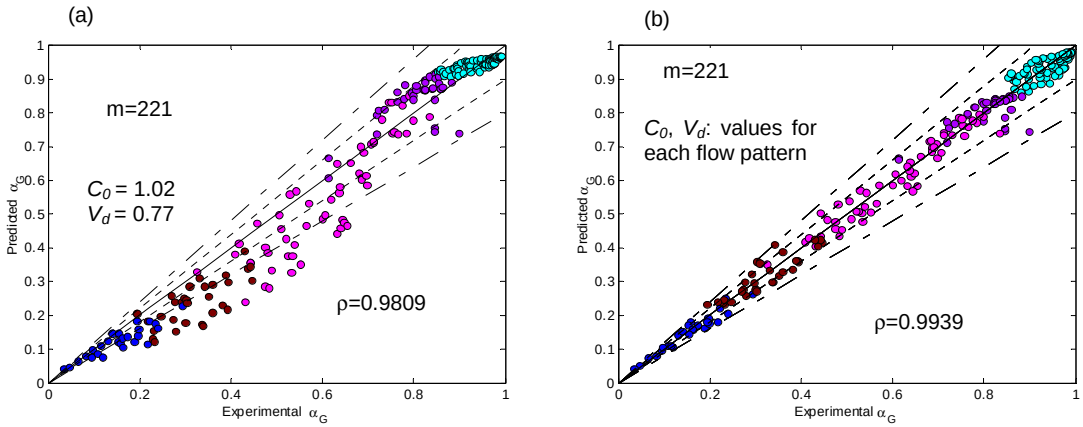


Figure 4-10: Prediction results using E_{V_G} (Data: Spedding & Nguyen (1976), air-water system, vertical flow, D=1.79 inch)

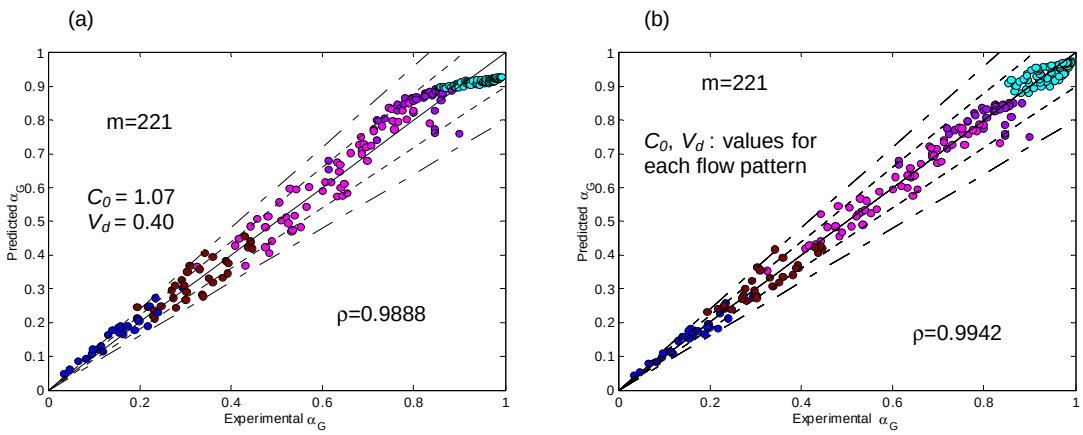


Figure 4-11: Prediction results using E_{α_G} (Data: Spedding & Nguyen (1976), air-water system, vertical flow, D=1.79 inch)

4.3.2. Incorporation of α_G into Correlations of C_0 and V_d

We represent C_0 and V_d as functions of α_G. As a first approximation, the following linear functions are considered:

$$\begin{aligned} C_0 &= a\alpha_G + b \\ V_d &= c\alpha_G + d \end{aligned} . \tag{4-5}$$

where a, b, c and d are constants.

Substituting Eq. (4-5) into Eq. (4-4), we get the following objective function:

$$E_{\alpha_G} = \sum_{i=1}^{m} \left[\alpha_{Gi} - \alpha_{Gi}^{*} \right]^2 , \tag{4-6}$$

where

$$\alpha_{Gi}^{*} = \frac{V_{SGi}}{\left(a\alpha_{Gi}^{*} + b\right)V_{Mi} + \left(c\alpha_{Gi}^{*} + d\right)} . \tag{4-7}$$

We note that Eq. (4-7) is implicit in α_G. However, since a linear expression for C_0 and V_d is assumed, the estimated α_G can be expressed explicitly. Therefore, the solution procedure for Eq. (4-6) is still a standard nonlinear least square problem, and the same Gauss Newton algorithm applied in the previous section is used.

The results using this method are shown in Figs. 4-12 and 4-13. Compared to Fig. 4-9 (a), we get a better prediction for α_G in Fig. 4-12, especially in the annular-mist flow regime (α_G near 1). For another data set, the same observation can be made (compare Figs. 4-11 (a) and 4-13), though the improvement here is not as significant as in the previous case.

We have shown that when the flow pattern information is considered, E_{V_G} and E_{α_G} can provide us with equivalent predictions (Figs. 4-10(b) and 4-11(b)). So it is worthwhile to consider using Eq. (4-5) in Eq. (4-3), to account for the effects of α_G when using E_{V_G} as the objective function. The advantage of E_{V_G} is the linearity in the resulting least square problem and the explicit expression for the estimated V_G. This will allow for the use of simpler algorithms when a more complicated form for C_0 and V_d is used. Unfortunately, as shown in Figure 4-14, the combination of the linear form for C_0 and V_d and the objective function of E_{V_G} does not give us results that are as accurate as those in Fig. 4-13. This demonstrates that there is a clear advantage in using E_{α_G} rather than E_{V_G}.

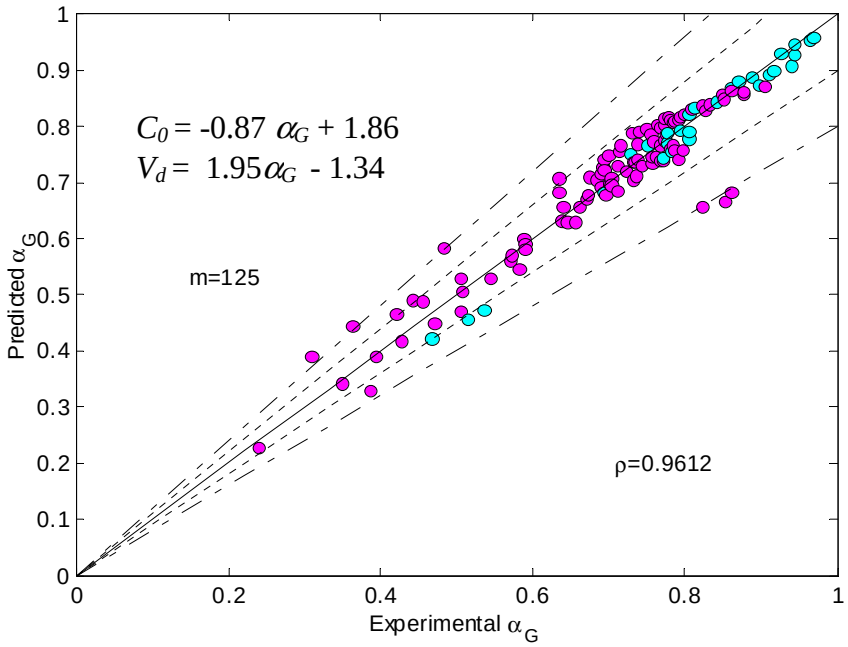


Figure 4-12: Prediction result using new approach (Data: SU66, Govier *et al.* (1957), air-water system, vertical flow, *D*=1.02 inch)

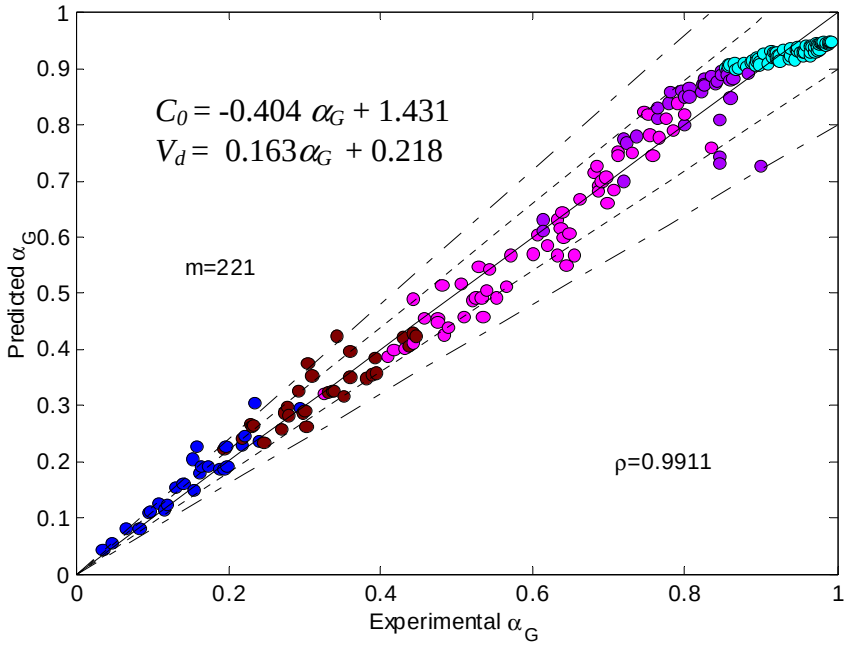


Figure 4-13: Prediction result using new approach (Data: Spedding & Nguyen (1976), air-water system, vertical flow, *D*=1.79 inch)

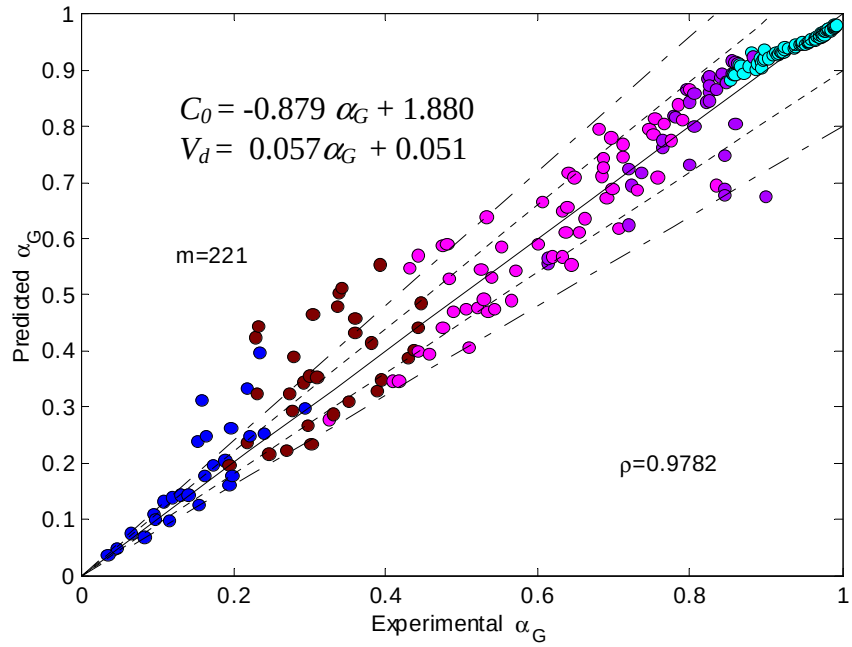


Figure 4-14: Prediction result using E_{V_G} and linear form of C_0 and V_d (Data: Spedding & Nguyen (1976), air-water system, vertical flow, D=1.79 inch)

4.4. Application of Proposed Method to Other Inclination Angles

So far, we have investigated only vertical flow. Earlier in this chapter, however, we showed that the drift-flux model is also applicable for horizontal flow. In this section, we will therefore apply the method described above to flows in other inclination angles. The objective is to assess the applicability of the drift-flux model to flow in other inclinations, as well as to test the robustness of the method we proposed for the determination of C_0 and V_d.

The results for upward and downward flows at various inclination angles (θ), using the data of Spedding & Nguyen (1976), are displayed in Fig. 4-15. For the upward flow (θ=70°, 45° and 21°), we obtain high correlation coefficients and the predictions are comparable to those for vertical flow. For near-horizontal and downward flows, the data shows significant scatter. In downward flow, stratified flow takes the place of intermittent flow. Thus few data points are in the region of low gas volume fraction. The results for horizontal flow will be shown later.

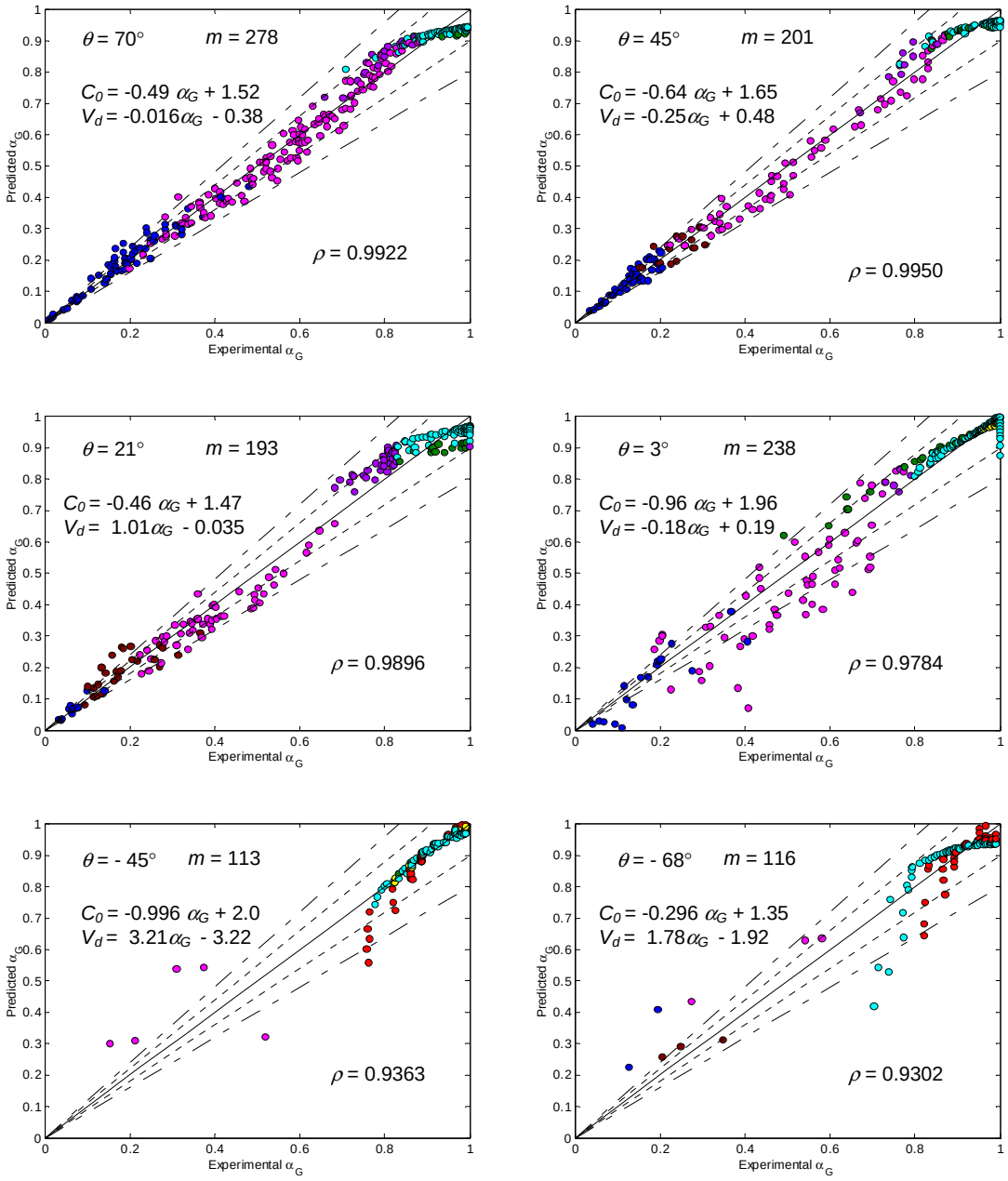


Figure 4-15: Prediction results for other inclination angles (Data: Spedding & Nguyen (1976), air-water system, *D*=1.79 inch)

Fig. 4-16 shows results using other data sets in the Stanford Multiphase Flow Database. These data are all from the same source (Mukherjee, 1979), but the pipe diameter ranges from 1-2 inches. Again, the results for downward flow are not as good as those for upward flow.

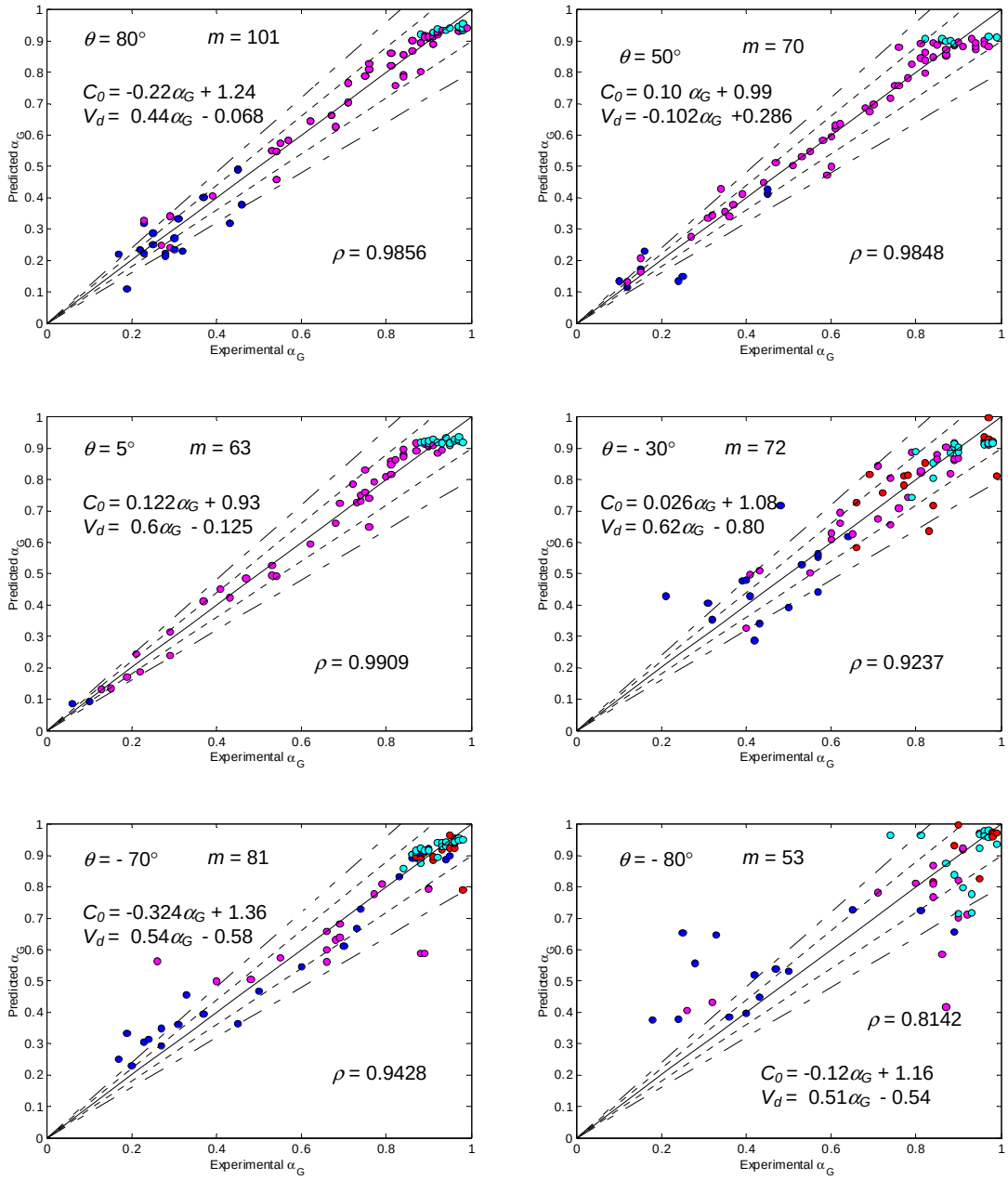


Figure 4-16: Prediction results for other inclination angles (Data: SU175-SU198: Mukherjee, 1979)

We next consider horizontal flow. In Figs. 4-17(a) and 4-18(a), we show the results for the case with C_0 and V_d constant. Figs. 4-17(b) and 4-18(b) display the predictions for α_G when C_0 and V_d are estimated in each flow pattern. This data and the data for vertical flow presented in Figs. 4-10 and 4-11 are from the same researchers (Spedding &

Nguyen, 1976). However, the results for horizontal flow are much less accurate than those for vertical flow. The data points in the stratified flow regime (represented by red points) show more scatter (in Fig. 4-17(a), they are toward the lower right of the figure), though a very high correlation between V_G and V_M was achieved in Fig. 4-2(a). When we apply the linear forms of C_0 and V_d to this data set, the nonlinear least square algorithm may converge to unphysical (complex) values for C_0 and V_d. In this case, additional constraints must be introduced into the optimization procedure. This problem has been observed for some horizontal and downward flows, but not for upward flows.

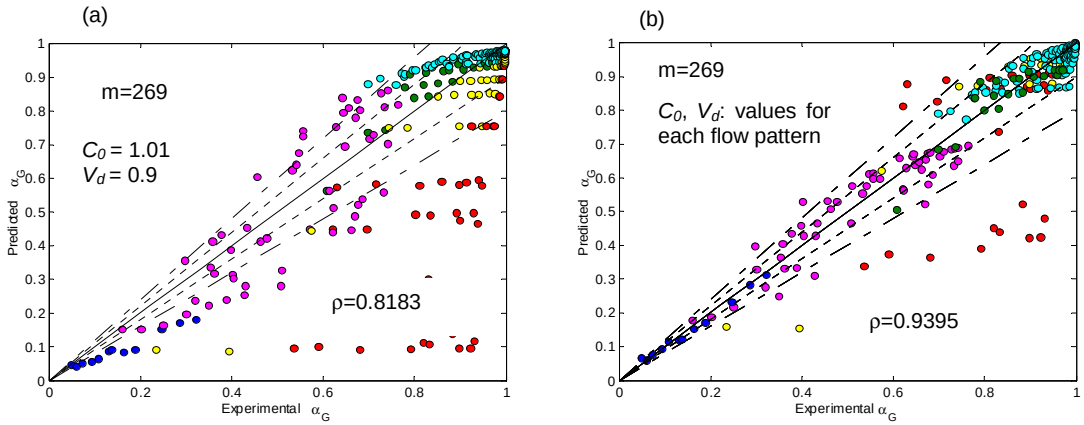


Figure 4-17: Prediction results using E_{V_G} (Data: Spedding & Nguyen (1976), air-water system, horizontal flow, D=1.79 inch)

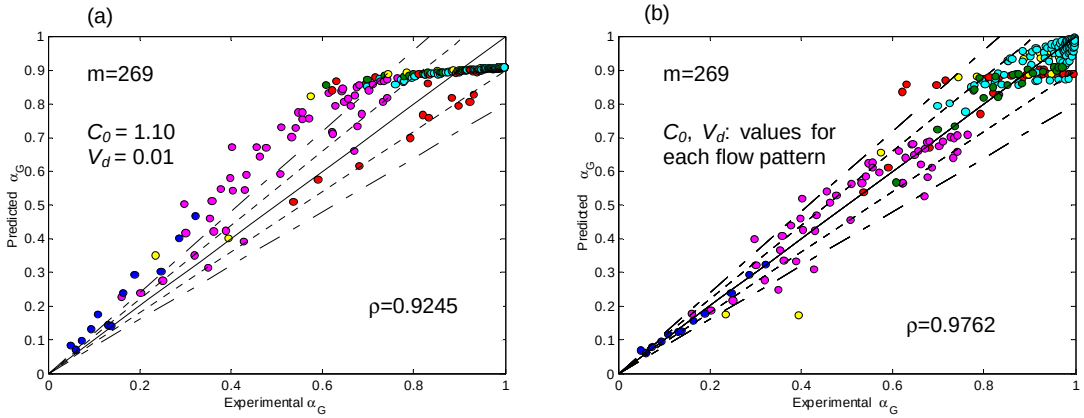


Figure 4-18: Prediction results using E_{α_G} (Data: Spedding & Nguyen (1976), air-water system, horizontal flow, D=1.79 inch)

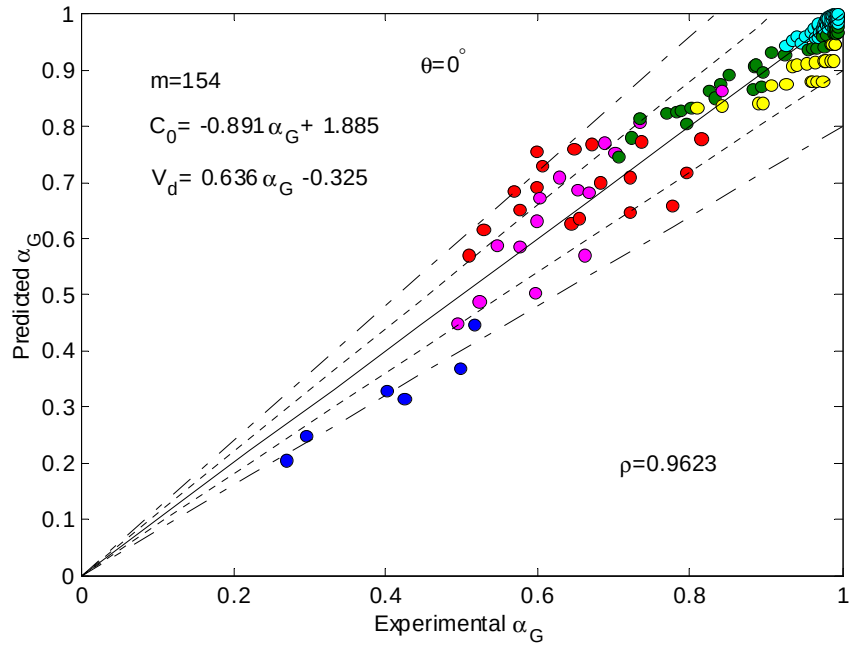


Figure 4-19: Prediction result for horizontal flow (Data: Chen & Spedding (1979), air-water system, *D*=1.79 inch)

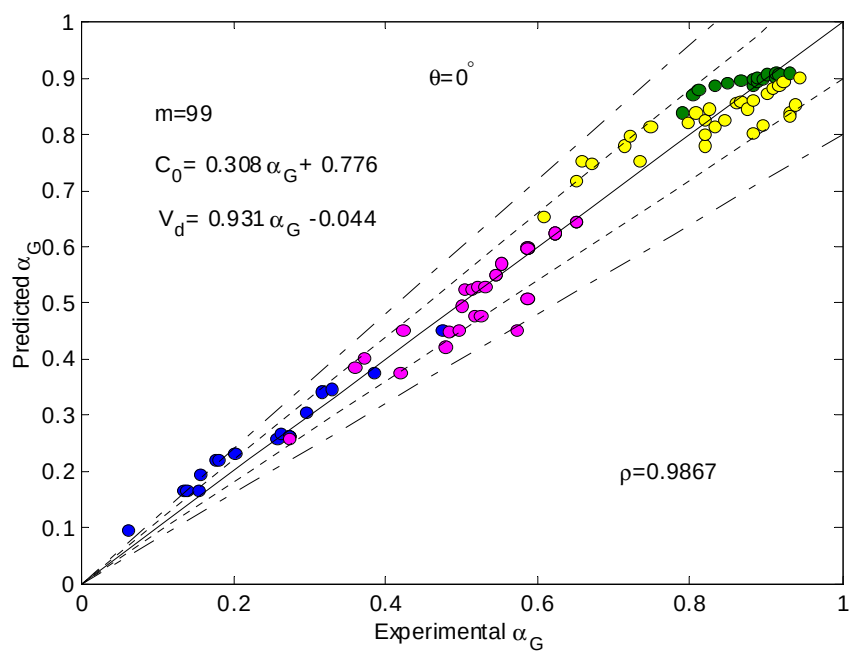


Figure 4-20: Prediction result for horizontal flow (Data: Franca & Lahey (1992), air-water system, *D*=0.75 inch)

For horizontal flow, good predictions using our method can be achieved for some data sets (see Figs. 4-19 and 4-20). However, the overall performance of the method for downward flow and horizontal flow is not satisfactory, compared to that achieved for vertical and upward flows. The determination of drift-flux parameters for downward and horizontal flow still needs further investigation.

4.5. Discussion of Drift Velocity V_d

We proposed a method for the determination of drift-flux parameters that works well for upward flows. So far, we have only applied these parameters to the data set for which they were generated. To obtain a more general correlation, we need to apply the procedure to multiple data sets. However, before we move on to this step, it is useful to consider the physical meanings of C_0 and V_d. Is the linear model we assumed appropriate and adequate? Is the behavior we get here consistent with the physical observations? We now consider these issues.

4.5.1. Physical Meaning of C_0 and V_d

For most of the data sets, C_0 and V_d display behavior similar to that shown in Fig. 4-21. C_0 decreases to 1.0 as α_G approaches 1.0, while V_d increases with α_G. As we have explained, C_0 accounts for the effects of the non-uniform distribution of both velocity and concentration profiles. As α_G approaches 1.0, which means high gas volume fraction and high flow rates, the profiles tend to distribute uniformly, so $C_0 \sim 1$. When α_G approaches zero, non-uniform effects are stronger so C_0 deviates from one.

V_d accounts for the local relative velocity between the two phases. A limiting case can be thought of as a single gas bubble rising through a liquid. This single bubble rise velocity is also called terminal gas rise velocity, designated V_∞. It can be calculated as follows, as determined from experimental results (Zuber & Findlay, 1965):

$$\text{in bubbly flow regime:} \quad V_\infty = 1.53\left[\frac{\sigma g(\rho_L - \rho_G)}{\rho_L^2}\right]^{\frac{1}{4}}, \tag{4-8}$$

and in slug flow regime: $V_\infty = 0.35\left[\dfrac{g(\rho_L - \rho_G)D}{\rho_L}\right]^{\frac{1}{2}}$. (4-9)

For the air-water system of Fig. 4-21, this velocity is 0.25m/s (Eq. (4-8)) or 0.23 m/s (Eq. (4-9)). Our prediction (0.218 m/s) in this limit ($\alpha_G \to 0$) is close to these values. The problem lies in the region of high gas volume fraction. Due to the effect of swarms of bubbles, we expect V_d to go to zero when α_G is one. However, the opposite trend is obtained in Fig. 4-21 (b).

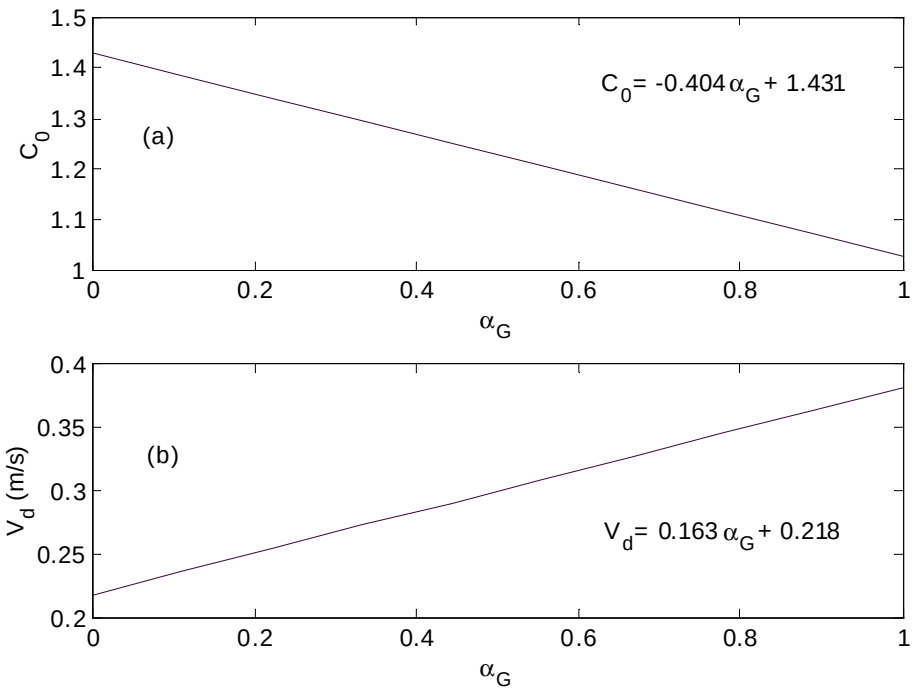


Figure 4-21: Typical behavior of calculated C_0 and V_d (Data: Spedding & Nguyen (1976), air-water system, vertical flow, D=1.79 inch)

Flores *et al.* (1998) studied the drift-flux model in oil-water flow and expressed the drift velocity as a function of oil volume fraction. An expression of the same form can also be used in the gas-liquid flow* (Gomez *et al.*, 2000):

* A similar form is also presented by Zuber & Findlay (1965). Their value for the power ranges from 0 to 3.

$$V_d = V_\infty (1-\alpha_G)^{0.5}. \tag{4-10}$$

The interesting point here is the trend of V_d at high α_G. This expression also indicates that $V_d \rightarrow 0$ as $\alpha_G \rightarrow 1.0$. Our simple linear model does not reproduce this behavior in that limit. Further investigation of V_d in high volume fractions is therefore needed.

4.5.2. Further Investigation of V_d as $\alpha_G \rightarrow 1$

Using the same data set, instead of applying our procedure to the entire range of volume fractions, we now consider only the data at high α_G. The results for C_0 and V_d when $\alpha_G \subset [0.95, 1.0]$ are shown in Fig. 4-22. C_0 is seen to decrease from 1.05 to 1.0, which is consistent with our previous observation. In this range of α_G, V_d now trends toward zero, in contrast to the previous increasing trend. The comparison between the calculated and experimental α_G displays a very high correlation. This shows that, although our previous linear model for the entire range of α_G provided good global accuracy, it failed to capture the correct behavior in limit $\alpha_G \rightarrow 1.0$.

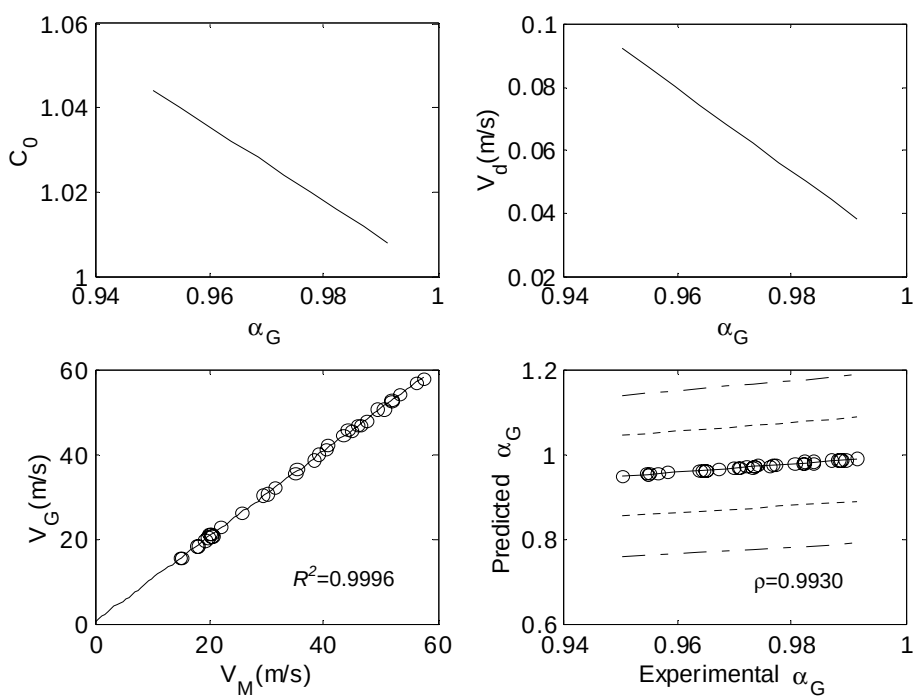


Figure 4-22: Behavior of C_0 and V_d at high $\alpha_G \subset [0.95, 1.0]$ (Data: Spedding & Nguyen (1976), air-water system, vertical flow, D=1.79 inch)

We now apply the same procedure to other inclination angles. In Fig. 4-23, similar behavior is observed for upward and horizontal flows. However, we cannot infer from this data where the inflection point in V_d should be --- for some data sets, we observe a decreasing trend only for $\alpha_G > 0.95$, while for other data sets, this behavior is observed for $\alpha_G > 0.8$.

Fig. 4-24 illustrates our results for downward flows. The trends are similar to those observed in upward flow, though negative drift velocities occur for downward flow. This is expected since drift velocity can be thought of as gas rise velocity.

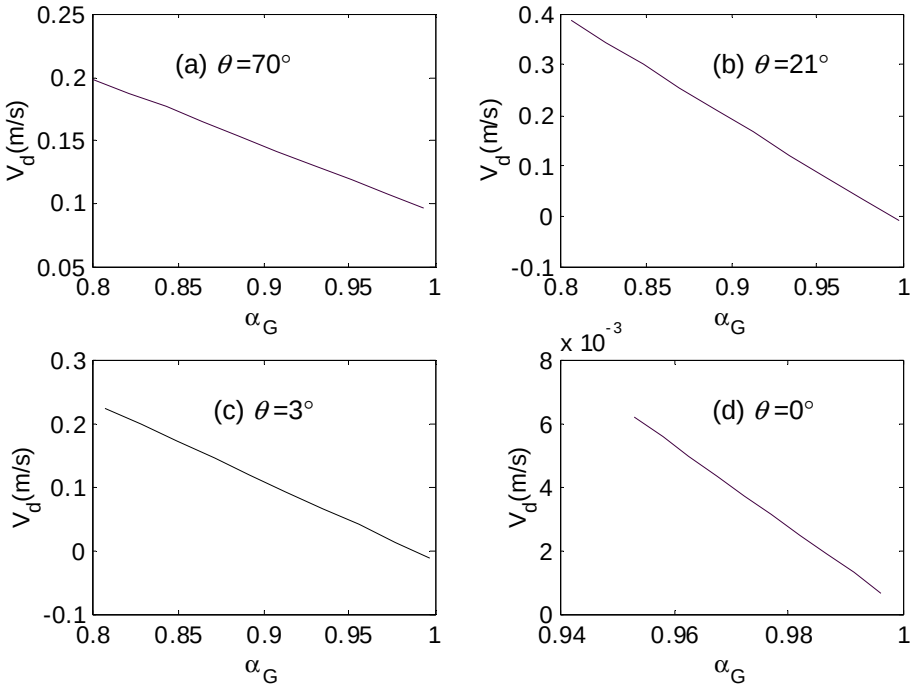


Figure 4-23: Behavior of V_d at high α_G for upward and horizontal flows (Data: Spedding & Nguyen (1976), air-water system, D=1.79 inch)

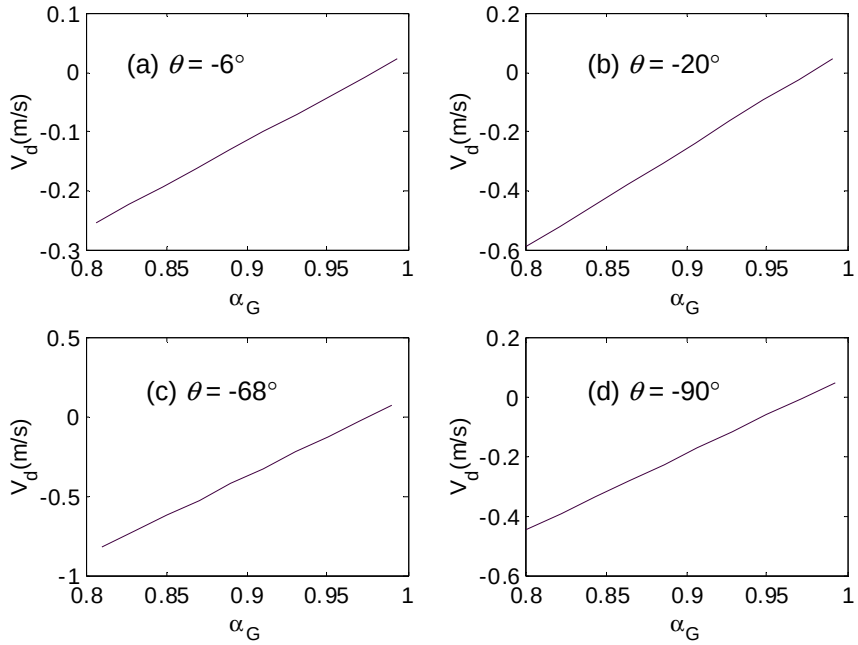


Figure 4-24: Behavior of V_d at high α_G for downward flows (Data: Spedding & Nguyen (1976), air-water system, D=1.79 inch)

4.5.3. Effects of Inclination Angles on V_d

It is important to be able to predict the effect of pipe inclination angle on the drift velocity. We demonstrated this effect in the range of high gas volume fractions in the previous section. Hasan & Kabir (1999) developed a formula for the terminal gas rise velocity for oil-water flow, which is valid for upward flow for $\theta' \leq 70°$ (θ' is measured from vertical):

$$V_{\infty}^{\theta'} = V_{\infty}\sqrt{\cos\theta'}\left(1+\sin\theta'\right)^2, \tag{4-11}$$

where V_{∞} is the terminal gas rise velocity in vertical upward flow. We introduce a simple extension of Hasan & Kabir's correlation for downward flow:

$$V_{\infty\theta'} = -V_{\infty}\sqrt{-\cos\theta'}\left(1+\sin\theta'\right)^2, \tag{4-12}$$

since a negative drift velocity is expected in downward flow.

In this work, we use the data of Spedding & Nguyen (1976) to compute the drift velocities in each inclination. The dependency of V_d on inclination is shown in Fig. 4-25.

Note that Hasan & Kabir's correlation is for V_∞, that is V_d as α_G approaches zero. Our calculations use the data over the whole range of α_G. However, our result displays a trend similar to that of Eq. (4-11). For downward flow, as discussed above, the fitting between the predicted and experimental α_G is not very accurate so our parameters may be only approximate in this range.

A more accurate way of investigating the influence of inclination is to calculate V_d in several ranges of gas volume fraction, as shown in Fig. 4-26. In downward flow, the intermittent flow regime is replaced by stratified flow, so only high gas volume fractions are observed in downward flow. One observation we can make from Fig. 4-26 is that for upward flow, the drift velocity does not depend strongly on α_G.

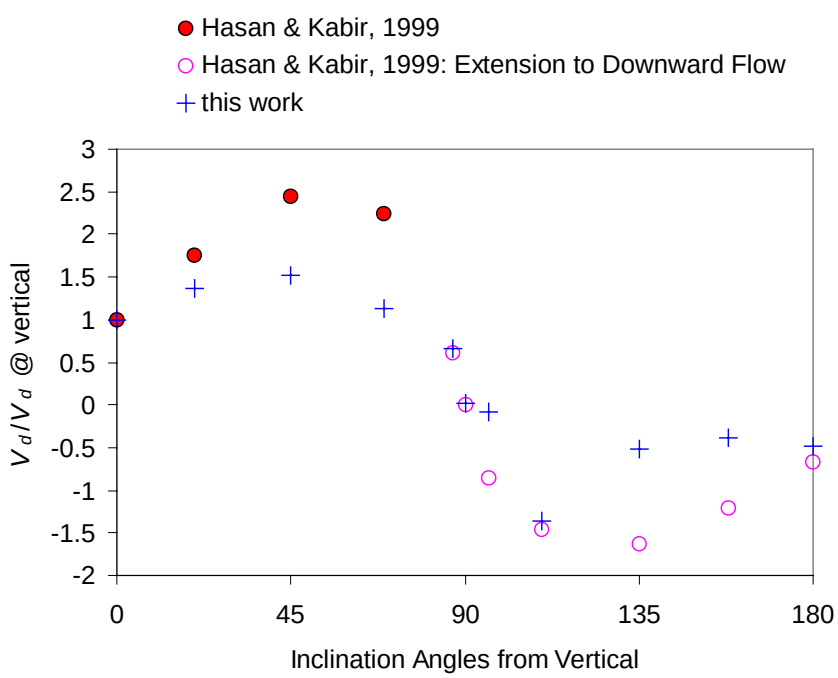


Figure 4-25: V_d in different inclination angles (Data: Spedding & Nguyen (1976), air-water system, D=1.79 inch)

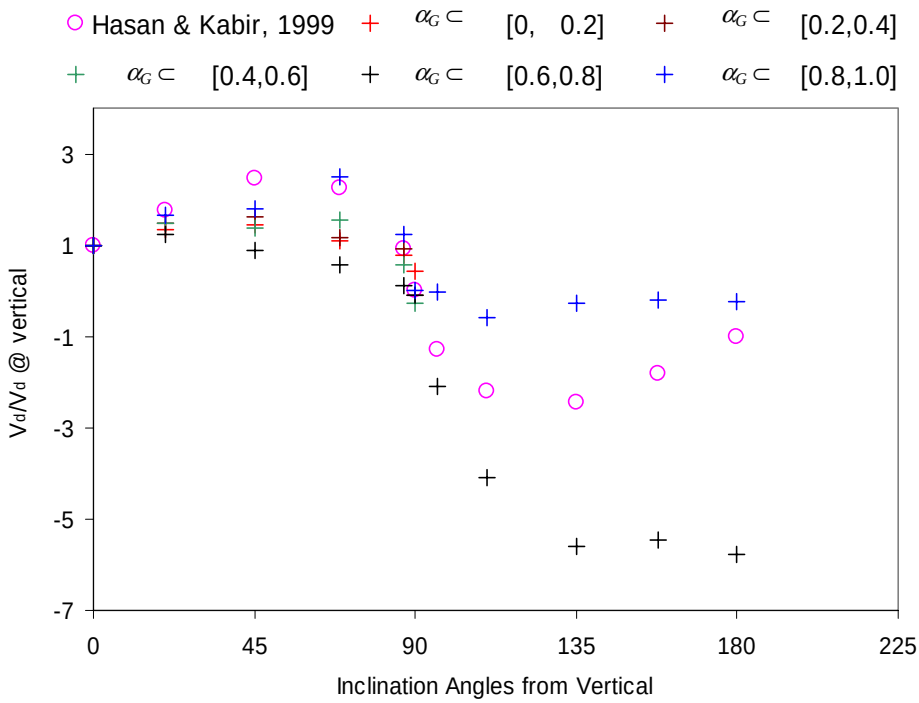


Figure 4-26: V_d in different ranges of α_G in different inclination angles (Data: Spedding & Nguyen (1976), air-water system, D=1.79 inch)

4.6. Evaluation of the Drift-Flux Model in Eclipse

The drift-flux model is used widely in reservoir simulators when the wellbore flow is coupled with the reservoir flow. One example is in the Multi-Segment Well calculations (Schlumberger GeoQuest, 2000) in Eclipse. Here, we use our data to evaluate the drift-flux model in Eclipse. In the Eclipse formulation, C_0 depends not only on α_G, but also on V_M. V_d is taken to be a function of α_G, V_M and θ. The detailed description for the development and formulation can be found in the Eclipse reference manual (Schlumberger GeoQuest, 2000). Recall that our development only considers the dependency of the parameters on α_G.

In Fig. 4-27, we plot the curve C_0 vs. α_G using the Eclipse model. The lines with different colors correspond to different mixture velocities V_M. In Fig. 4-27(a), we see the mixture velocity has a very strong effect on the value of C_0. For example, when α_G is equal to 0.6, increasing V_M to 20 m/s makes C_0 equal to 1.0. Although the velocity and concentration

profiles tend to distribute more uniformly at high flow rates, the high gas volume fraction appears to have a stronger impact on the distribution profile. This is illustrated by the value of C_0 calculated from our data, shown as points in Fig. 4-27. For different ranges of α_G, a constant value for C_0 can be obtained, and the average α_G and average V_M can also be calculated. For the values for C_0 associated with the three highest V_M (19.7 m/s, 25.0 m/s and 38.6 m/s), we obtain C_0 greater than 1 except when α_G is extremely close to 1. It is demonstrated that $C_0 \rightarrow 1$ only when $\alpha_G \rightarrow 1$, This is in contrast to the Eclipse model, which shows a strong dependency on V_M that forces the C_0 curves to approach one quickly.

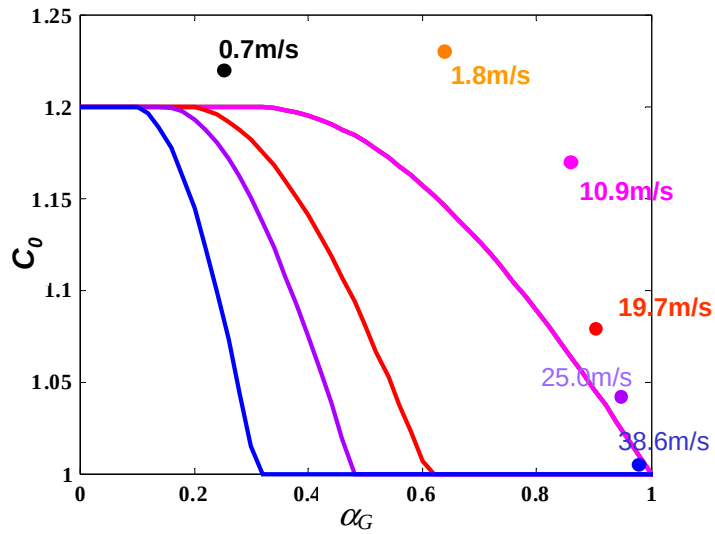
There are three user-definable parameters in the Eclipse model (A, B and F_v) that can be used to adjust the value of C_0. The parameter A is the value of C_0 at low values of α_G and V_M. B is the value of the gas volume fraction at which C_0 will reduce from the value A. F_v adjusts the sensitivity of the C_0 curve to the mixture velocity. In Fig. 4-27(b), we tune these parameters to obtain a better match. Essentially, we modify the parameters to reduce the effect of V_M. The five curves corresponding to different mixture velocities now collapse to a single curve. This shows that the impact of the mixture velocity is overestimated for this data. However, it should be noted that this data is for pipes of diameter 1-2 inches. It is possible that V_M will have a greater effect on C_0 for larger diameter pipe.

The relationship between α_G and V_M is demonstrated in Fig. 4-28, which is obtained in a similar manner as Fig. 4-4 --- the average mixture velocity is computed in each flow pattern. We see that the high volume fraction corresponds to high mixture velocity. This relation is also displayed clearly in the flow pattern map in Fig. 4-7 --- the annular-mist flow cannot be expected to occur in low flow rates. Therefore, α_G can be expected to represent V_M, suggesting that further dependency on V_M is not required.

Fig. 4-29 displays the correlation for V_d used in Eclipse. The correction for θ dependency is achieved using Hasan & Kabir's formula (Eq. (4-11)). The behavior of V_d at the limiting values of α_G ($\alpha_G \rightarrow 0$ and $\alpha_G \rightarrow 1$) is consistent with our observations. The V_d

curve in Eclipse provides an alternative when a more complicated model is required, rather than our simple linear model for V_d.

(a) A=1.2, B=0.3, F_v=1.0 (Default Values in Eclipse)



(b) A=1.225, B=0.4, F_v=0.6 (Modified Values in Eclipse)

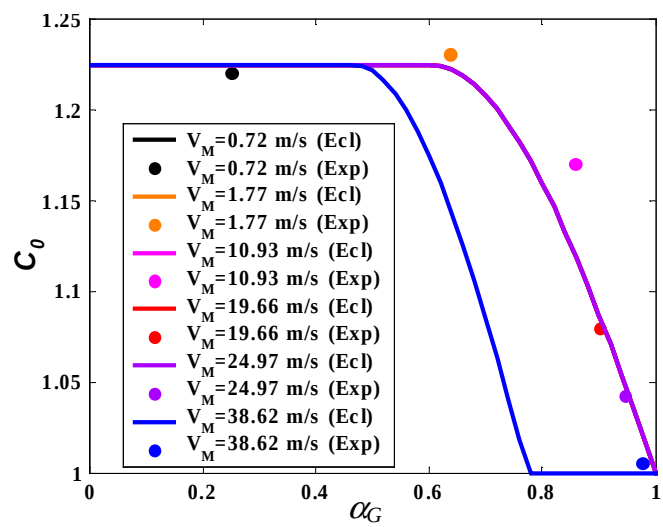


Figure 4-27: Comparison of C_0 between Eclipse correlation and calculated values for different V_M (Data: Spedding & Nguyen (1976), air-water system, vertical flow D=1.79 inch)

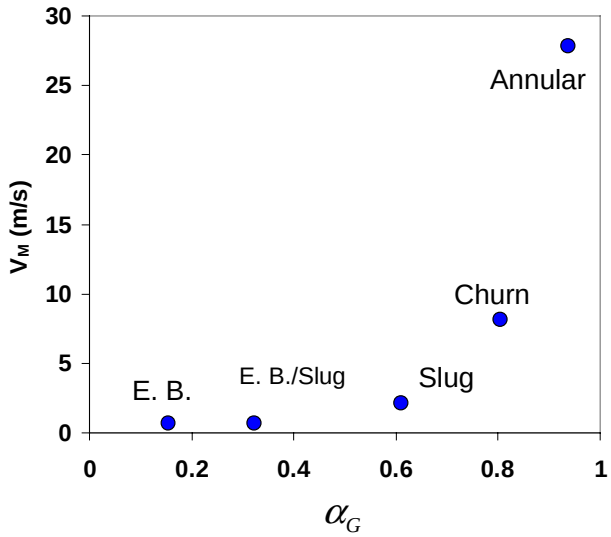


Figure 4-28: Relation between α_G and V_M (Data: Spedding & Nguyen (1976), air-water system, vertical flow D=1.79 inch)

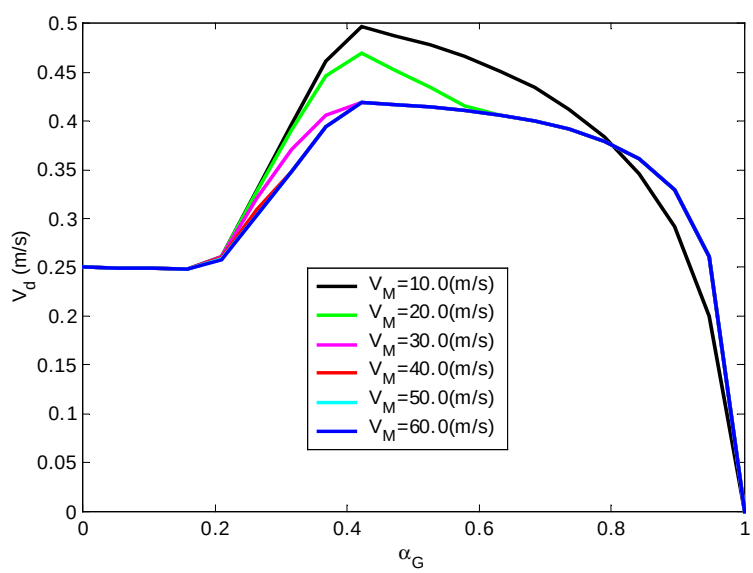


Figure 4-29: V_d curve in Eclipse (air-water system, D=1.79 inch)

We now evaluate the correlation in Eclipse using our current data. The performance is shown in Fig. 4-30. Fig. 4-30 (a) displays the comparison using default values in Eclipse, while Fig. 4-30 (b) is that for the modified values suggested above. A noticeable improvement is obtained at high gas volume fractions. This is due to the correction of C_0 at high α_G, as indicated in Fig. 4-27.

In this chapter, we have shown that the drift-flux model parameters (C_0 and V_d) have a strong dependency on flow regimes. It was also illustrated that α_G is a good indicator of the flow pattern. Based on this observation, a method was proposed to model the two parameters C_0 and V_d as functions of α_G. Although satisfactory prediction results were obtained, our linear model for V_d does not incorporate the physical behavior of V_d over the entire range of α_G. Using our data, the drift-flux model used in Eclipse was evaluated. We found that the effects of the mixture velocity V_M are overestimated in the model. A modification was suggested to reduce the effects of V_M, and the resulting predictions show an improvement at high gas volume fraction.

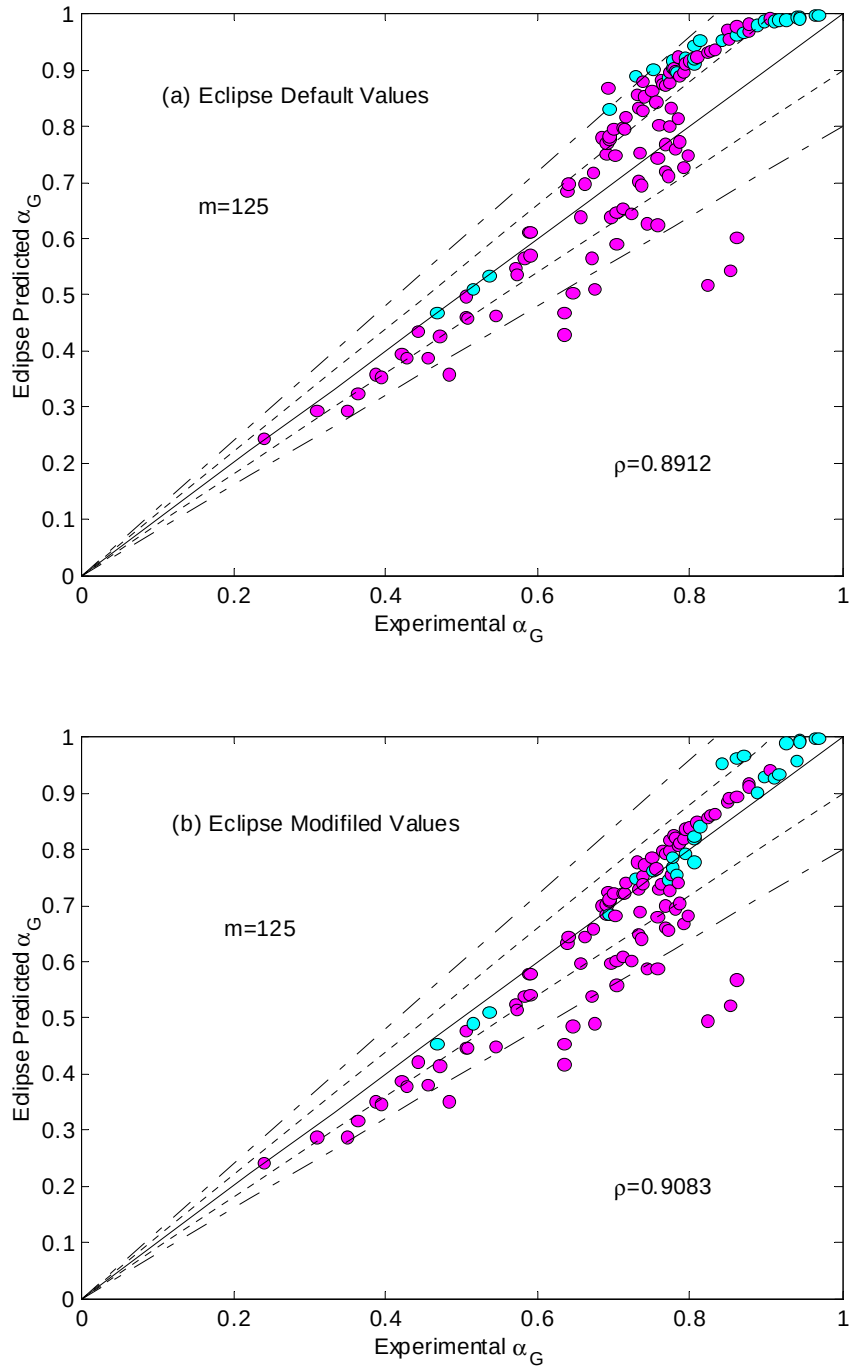


Figure 4-30: Performance of the drift-flux model correlation in Eclipse (Data: SU66, Govier *et al.* (1957), air-water system, vertical flow, *D*=1.02 inch)

Chapter 5

5. Conclusions and Future Work

5.1. Summary and Conclusions

This work addressed the modeling of two-phase gas liquid flow in pipes. It included discussions of the flow pattern transition predictions in mechanistic models, and a detailed determination of the drift-flux model parameters.

The Petalas & Aziz (1998) mechanistic model was evaluated in terms of its ability to predict flow pattern transitions. Based on the data considered, this model provided acceptable transition predictions and exhibited reasonable trends for fluid property variations. However, the transition to dispersed bubble flow and the transition between stratified flow and intermittent flow can be improved.

Other transition criteria were compared with experimental data. The mechanism for the transition to dispersed bubble flow is considered to be the turbulent fluctuation forces breaking up the gas phase into dispersed bubbles. Among the models developed from this mechanism, Barnea's (1986) model gave the most accurate result. The Petalas & Aziz (1998) model is based on the liquid volume fraction in the slug. It provides correct trends, but the critical value used in the model requires some tuning.

For the transition from intermittent flow to annular-mist flow, the holdup based transition criterion ($\alpha_G > 0.75$) can provide reasonable transition boundaries. Its accuracy, however, may rely on the correlations used to calculate the in situ gas volume fraction. The same kind of problem was investigated in the transitions for stratified flow. Although, a Kelvin-Helmholtz wave stability analysis is well-established for this transition, we showed that the interfacial friction factor has a strong effect on the transition. The simple approximation $f_i=f_{wG}$ is recommended for the flow pattern prediction.

The drift-flux model is a simple but useful way of calculating the holdup. The use of appropriate values for C_0 and V_d determines the accuracy of this model. It was observed that this model may be applied to all flow patterns, and that flow patterns have a strong impact on the values for C_0 and V_d. A method was proposed to develop correlations for C_0 and V_d in which both C_0 and V_d are functions of α_G. This is reasonable since α_G was shown to approximately represent the flow pattern information. The two parameters were determined by directly minimizing the errors between the experimental and the estimated α_G. Compared to the previous approach, minimizing the objective function of the actual gas velocity, significant improvement was achieved. The performance of the drift-flux correlations in Eclipse was also evaluated using the current data. We showed that the effect of the mixture velocity is overestimated in the Eclipse model, and specific user-definable parameters were suggested to improve the prediction of α_G at high gas volume fraction.

The resulting C_0, which has a value close to 1 at high α_G, is consistent with its physical meaning. However, our simple linear model for V_d does not account for the physical behavior of V_d as $\alpha_G \rightarrow 1$. A more realistic model is needed to capture behavior in this limit correctly.

5.2. Recommendations for Future Work

The Stanford Multiphase Flow Database contains mainly data on air-water systems in small (1-2 inch) diameter pipes. The extension of the experimental database to other fluids and larger pipes will allow us to extend our models to a wider range of conditions.

The development of new models should be based on the actual mechanisms behind the flow pattern transitions. Although the effects of fluid properties were shown in this work not to be as significant as that of inclination angle, the examination of fluid property variations can be a way of investigating new mechanisms. In both Chapter 3 and Chapter 4, we showed that volume fractions indicate flow patterns. Therefore, the holdup based transition prediction is reasonable. This may allow us to eliminate discontinuities in holdup calculations in the mechanistic model.

As shown, even in a mechanistic model, a large number of empirical correlations are required. The accuracy of these correlations will affect the performance of the mechanistic model. However, future research in this field should not entail only the addition of new correlations to this already crowded list. Rather, new correlations should be based on clear underlying physical phenomena.

The extension of drift-flux modeling is a promising research area. The drift-flux model needs to be further assessed for horizontal flow and downward flow, where the stratified flow pattern occurs. The method proposed in this work can be applied to a variety of other conditions. However, before it is generalized to wider conditions, the behavior of V_d as $\alpha_G \rightarrow 1$ should be further studied. The linear model should also be extended --- one way of doing this is to apply the procedure developed here in different ranges of α_G. The flow pattern information could be used to determine the appropriate ranges. The dependency of V_d on θ is another problem that deserves further attention. Improvement of the drift-flux model along the lines described here will ultimately result in more accurate well models in reservoir simulators.

Nomenclature

A	=	Pipe cross sectional area
C	=	Input volume fraction
C_0	=	Distribution parameter, profile parameter
d	=	Size of gas bubble
d_b	=	Bubble diameter
D	=	Pipe internal diameter
E_S	=	Surface free energy due to surface tension
E_T	=	Turbulent kinetic energy
f	=	Friction factor
f_{wG}	=	Gas/wall friction factor
f_{wL}	=	Liquid/wall friction factor
F_B	=	Buoyant force
F_r	=	Froude number
F_T	=	Turbulent force
g	=	Gravitational acceleration
h_L	=	Liquid height
p	=	Pressure
Q	=	Volumetric flow rate
Re	=	Reynolds number
S	=	Pipe perimeter
v'	=	Fluctuation of velocity
V_d	=	Drift velocity of gas
V_G	=	Actual gas velocity
V_L	=	Actual liquid velocity
V_{SG}	=	Superficial gas velocity
V_{SL}	=	Superficial liquid velocity
$V_{SL,\delta}$	=	Superficial velocity of the liquid film

V_M	=	Volumetric flux of the mixture
V_∞	=	Terminal gas rise velocity in vertical flow
$V_\infty^{\theta'}$	=	Terminal gas rise velocity in inclined pipe
x	=	Axial coordinate in pipe

Greek Letters

α	=	In situ volume fraction
$\alpha_{L,s}$	=	Liquid volume fraction in the liquid slug
ε	=	Dissipation rate of turbulent kinetic energy
θ	=	Pipe inclination angle (measured from horizontal)
θ'	=	Pipe inclination angle (measured from vertical)
μ	=	Dynamic fluid viscosity
ν	=	Kinematic fluid viscosity
ρ	=	Fluid density
σ	=	Interfacial tension/surface tension
τ_i	=	Interfacial friction shear stress
τ_{wG}	=	Gas/wall friction shear stress
τ_{wL}	=	Liquid/wall friction shear stress

Subscripts

G	=	Gas phase
L	=	Liquid phase
i	=	Interfacial

References

Arif, H. "Application of Computational Fluid Dynamics to the Modeling of Flow in Horizontal Wells". M.S. Report, Stanford University, Stanford, CA, 1999.

Ansari, A. M., Sylvester, N. D., Sarica, C., Shoham, O. and Brill, J. P. "A Comprehensive Mechanistic Model for Upward Two-Phase Flow in Wellbores". *SPE Production & Facilities*, May: 143-152, 1994.

Baker, A., Nielsen, K. and Gabb, A. "Pressure Loss, Liquid-Holdup Calculation Developed". *Oil & Gas J.*, March 14: 54-59, 1988.

Barnea, D. "Transition from Annular Flow and from Dispersed Bubble Flow --- Unified Models for the Whole Range of Pipe Inclinations". *Int. J. Multiphase Flow*, 12(5): 733-744, 1986.

Barnea, D. "A Unified Model for Predicting Flow-Pattern Transitions for the Whole Range of Pipe Inclinations". *Int. J. Multiphase Flow*, 13(1): 1-12, 1987.

Brill, J. P. and Arirachakaran, S. J. "State of the Art in Multiphase Flow". *Journal of Petroleum Technology*, May: 538-541, 1992.

Chen, J. J. J. and Spedding, P. L. "Data on Holdup, Pressure Loss and Flow Pattern in a Horizontal Pipe". Report 214, University of Auckland, Auckland, New Zealand, 1979.

Chen, X. T., Cai, X. D. and Brill, J. P. "A General Model for Transition to Dispersed Bubble Flow". *Chemical Engineering Science*, 52(23): 4373-4380, 1997.

Cheremisinoff, N. P. "*Encyclopedia of Fluid Mechanics, Volume 3, Gas-Liquid Flows*". Gulf Publishing Company, Houston, Texas, 1986.

Flores, J.G., Sarica, C., Chen, T.X. and Brill, J.P. "Investigation of Holdup and Pressure Drop Behavior for Oil-Water Flow in Vertical and Deviated Wells". *Journal of Energy Resources Technology, Transactions of the ASME*, 120: 8-14, 1998.

Franca, F. and Lahey Jr, R. T. “The Use of Drift-Flux Techniques for the Analysis of Horizontal Two-Phase Flows”. *Int. J. Multiphase Flow*, 18(6): 887-801, 1992.

Gomez, L. E., Shoham, O., Schmidt, Z., Chokshi, R. N. and Northug, T. “Unified Mechanistic Model for Steady-State Two-Phase Flow: Horizontal to Vertical Upward Flow”. *SPE Journal*, 5 (3): 339-350, 2000.

Govier, G. W. and Aziz, K. *“The Flow of Complex Mixtures in Pipes”*. Van Nostrand Reinhold Company, New York, NY, 1972.

Gregory, G. A., Nicholson, M. K. and Aziz, K. “Correlation of the Liquid Volume Fraction in the Slug for Horizontal Gas-Liquid Slug Flow”. *Int. J. Multiphase Flow*, 4(1): 33-39, 1978.

Hubbard, M. B. and Dukler, A. E. “The Characterization of Flow Regimes for Horizontal Two-Phase Flow”. *Proceedings of the 1966 Heat Trans. & Fluid Mech. Inst.*, Stanford U. Press, 1966.

Joseph, D. D., Bannwart, A. C. and Liu, Y. J. “Stability of Annular Flow and Slugging”. *Int. J. Multiphase Flow*, 22(6): 1247-1254, 1996.

Kaya, A. S., Chen, X. T., Sarica, C. and Brill, J. P. “Investigation of Transition from Annular to Intermittent Flow in Pipes”. *Proceedings of the 1999 ASME Energy Sources Technology Conference.* Houston, TX, February 1-3, 1999.

Kaya, A. S., Sarica, C. and Brill, J. P. “Comprehensive Mechanistic Model of Two-Phase Flow in Deviated Wells”. *SPE 56522*, SPE Annual Technical Conference and Exhibition, Houston, TX, October 3-6, 1999.

Kokal, S. L. and Stanislav, J. F. “An Experimental Study of Two-Phase Flow in Slightly Inclined Pipes --- I. Flow Patterns”. *Chemical Engineering Science*, 44(3): 665-679, 1987.

Li, Q., Yi, D. and Wang, N. *“Modern Numerical Analysis”* (in Chinese): 287-288. Higher Education Press, Beijing, China, 1995.

Mandhane, J. M., Gregory, G. A. and Aziz, K. “A Flow Pattern Map for Gas-Liquid Flow in Horizontal Pipes”. *Int. J. Multiphase Flow*, 1: 537-553, 1974.

McQuillan, K. W. and Whalley, P. B. “Flow Patterns in Vertical Two-Phase Flow”. *Int. J. Multiphase Flow*, 11(2): 161-175, 1985.

Mishima, K. and Ishii, M. “Flow Regime Transition Criteria for Upward Two-Phase Flow in Vertical Tubes”. *Int. J. Multiphase Flow*, 27(5): 723-737, 1984.

Newton, C. H. and Behnia, M. “Numerical Calculation of Turbulent Stratified Gas-Liquid Pipe Flows”. *Int. J. Multiphase Flow*, 26: 327-337, 2000.

Ouyang, L-B. “Single Phase and Multiphase Fluid Flow in Horizontal Wells”. Ph. D. thesis, Stanford University, Stanford, CA, 1998.

Ouyang, L-B. “Stratified Flow Model and Interfacial Friction Factor Correlations”. M.S. Report, Stanford University, Stanford, CA, 1995.

Petalas, N. and Aziz, K. “A Mechanistic Model for Multiphase Flow in Pipes”. CIM 98-39, Proceedings, 49th Annual Technical Meeting of the Petroleum Society of the CIM, Calgary, Alberta, Canada, June 8-10, 1998.

Petalas, N. and Aziz, K. “A Mechanistic Model for Stabilized Multiphase Flow in Pipes”. Stanford University, Stanford, CA, 1997.

Schlumberger GeoQuest. “Eclipse 200 Reference Manual – Multi-Segment Wells”, 2000 Release, 2000.

Shoham, O. “Flow Pattern Transitions and Characterization in Gas-Liquid Two Phase Flow in Inclined Pipes”. Ph. D. thesis, Tel-Aviv University, Ramat-Aviv, Israel, 1982.

Spedding, P. L. and Nguyen, V. T. “Data on Holdup, Pressure Loss and Flow Pattern for Two-Phase Air-Water Flow in an Inclined Pipe”. Report 122, University of Auckland, Auckland, New Zealand, 1976.

Taitel, Y. and Dukler, A. E. “A Model for Predicting Flow Regime Transitions in Horizontal and Near Horizontal Gas-Liquid Flow”. *AIChE J.* 22(1): 47-55, 1976.

Taitel, Y. “Advances in Two Phase Flow Mechanistic Modeling”. *SPE 27959*, 1995.

Taitel, Y., Barnea, D. and Dukler, A. E. “Modeling Flow Pattern Transitions for Steady Upward Gas-Liquid Flow in Vertical Tubes”. *AIChE J.* 26(3): 345-354, 1980.

Taitel, Y., Shoham, O. and Brill, J. P. “Simplified Transient Solution and Simulation of Two-Phase Flow in Pipelines”. *Chemical Engineering Science*, 44(6): 1353-1359, 1989.

Weisman, J., Duncan, D., Gibson, J. and Crawford, T. “Effects of Fluid Properties and Pipe Diameter on Two-Phase Flow Patterns in Horizontal Lines”. *Int. J. Multiphase Flow*, 5: 437-462, 1979.

Xiao, J. J., Shoham, O. and Brill, J. P. “A Comprehensive Mechanistic Model for Two-Phase Flow in Pipelines”. *SPE 20631*, SPE Annual Technical Conference and Exhibition, New Orleans, LA, September 23-25, 1990.

Zuber, N. and Findlay, J. A. “Average Volumetric Concentration in Two-Phase Flow Systems”. *Journal of Heat Transfer, Transactions of the ASME*, 87: 453-468, 1965.

Appendix A

A. Experimental Data

A.1 New Data Input

Table A-1 summarizes the data used that is not in the Stanford Multiphase Flow Database. Shoham's (1982) data and Kokal & Stanislav's (1987) data are used in the evaluation of transition models, while the data of Weisman *et al.* (1979) are used for the investigation of the influence of fluid properties on flow pattern transitions. We utilize data from Chen & Spedding (1979) and Franca & Lahey (1992) to test the correlations for drift flux model parameters.

Table A-1: Summary of New Data

Data Source	Inclination Angles	ID (Inches)	Gas-Liquid	Observation of		
				Flow Pattern	Holdup	Pressure Drop
Shoham (1982)	0, 10, -10, 90, -90	0.98, 2.01	Air-water	Yes	No	No
Chen & Spedding (1979)	0	1.79	Air-water	Yes	Yes	Yes
Franca & Lahey (1992)	0	0.75	Air-water	Yes	Yes	No
Kokal & Stanislay (1987)	0, 9, -9	1.02, 2.02 3.0	Air-water	Yes	No	No
Weisman *et al.* (1979)	0	1.0 2.0	*	Yes	No	No

Notation:

* Indicates several other fluids were used. See Table 3-1 for details.

A.2 Correction for Dataset SU199 ~ SU209

Datasets SU199~SU209 (data source: Spedding & Nguyen, 1976) in the Stanford Multiphase Flow Database cover the whole range of pipe inclinations and include information on pressure drop and holdup. Therefore, this is a valuable data source for model development and evaluation. However, we found some inconsistencies during our use of this data. Fig. A-1 shows an example of the flow pattern map for $\theta = 70^{\circ}$, in which some data points are reported as stratified flow. As discussed in Chapter 2, stratified flow does not exist when the inclination angle is above 15°. Instead, intermittent flow would be expected for upward flow.

Another error is the region of dispersed bubble flow. Although this flow pattern is found in a wide range of pipe inclinations, it usually occurs in the region of high liquid flow rates, where the turbulent fluctuations break up the bubbles. Apparently, the dispersed bubble flow in Fig. A-1 was mislabeled. Unlike the measurement of pressure drop and holdup, the determination of flow patterns is somewhat subjective. Although there are a lot of advances in the detection techniques of flow patterns, flow pattern information is usually determined through visual observation of the recorded video images. Different interpretations and different flow pattern terms may lead to some inconsistencies. Errors can also occur when the data are input to the database.

The original report (Spedding & Nguyen (1976)) was found and the data were corrected. The flow pattern determination in the original work was based on visual observations, with a tendency to include as many flow pattern descriptions as possible. There are 26 terms used for descriptions, among which 17 are the major ones. In Shoham's work (1982), he compared his observations with those of Spedding & Nguyen and combined some of the terms used. Based on the schematic pictures shown in the initial report and observations by Shoham, further combination (See Table A-2) was made to generate

reasonable flow pattern maps. The flow patterns described in Chapter 2 and some additional transitional flow patterns were used.

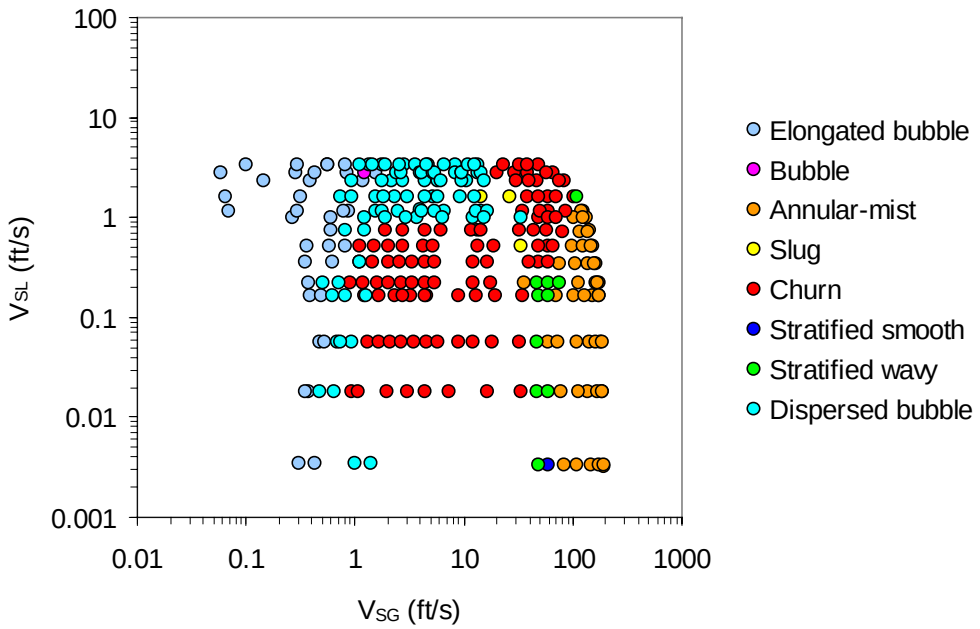


Figure A-1: Flow pattern map for Spedding & Nguyen (1976) data before correction (θ=70°, air-water, D=1.79 inch)

Table A-2: Combination of Flow Pattern Information for Datasets SU199-SU209

	Flow Patterns by Spedding & Nguyen (1976)	Flow Patterns by Shoham (1982)	Flow Patterns in this work
1	Stratified	Stratified smooth	Stratified smooth
2	Stratified + ripple	Stratified wavy	Stratified wavy
3	Stratified + inertial wave	Stratified wavy	Stratified wavy
4	Stratified + roll wave	Wavy annular	Wavy annular
5	Annular	Annular	Annular
6	Annular + roll wave	Annular	Annular
7	Annular + ripple	Annular	Annular

8	Droplet	Annular-mist	Annular-mist
9	Annular + blow through slug	Annular-mist	Annular-mist
10	Annular + droplet	Annular-mist	Annular-mist
11	Film + droplet	Annular-mist	Annular-mist
12	Pulsating froth	Churn	Churn
13	Annular +slug	Churn	Churn
14	Slug	Slug	Slug
15	Slug + froth blow through slug	Slug	Slug
16	Bubble	Slug	Slug
17	Bubble/Slug	Elongated bubble	Elongated bubble
18	Bubble/Froth	Elong-Bub/Slug	Elong-Bub/Slug
19	Slug/Annular	Elong-Bub/Churn	Elong-Bub/Slug
20	STR/BTS	Slug/Annular	Wavy annular
21	Slug/BTS	STR/Slug	Wavy annular
22	STR/Slug	Slug	Wavy annular
23	Slug/Bubble	STR/Slug	Wavy annular
24	Bubble/Droplet	Slug/Elong-Bub	Elong-Bub/Slug
25	STR/Droplet	Elong-Bub/Annular	Elong-Bub/Slug
26	Annular/IWA	STR/Annular	Wavy annular
27		Annular/STR	Annular/STR

Note: For the abbreviations used in the above table, refer to the report of Spedding & Nguyen (1976).

Some remarks about the above table:

- We grouped the flow patterns of Elongated bubble/Slug, Slug/Elongated bubble and Elongated bubble/Froth together into one flow pattern: **Elongated bubble/Slug** flow. The transition between slug flow and elongated bubble flow is not important, since the calculations of pressure drop and holdup in these two regimes are the same.

- There are few data points described as STR/Droplet, STR/BTS, Slug/Annular and STR/Annular. In this work, all these are interpreted as **Wavy Annular**, in which "most of the liquid flows as a film at the bottom of the pipe while aerated unstable waves are swept around the pipe periphery and wet the upper wall occasionally" (Shoham, 1982). This is a transitional zone among the annular-mist flow, stratified wavy flow and churn flow. Since the flow pattern transitions take place gradually, the existence of this transitional zone is reasonable.

- Even after our further combination, there are still some data points which appear to be in error. In upward flow of 21°, 45° and 70°, there are points designated stratified wavy flow. We changed these to wavy annular flow.

- For downward flow of -90°, the data in the original report is incomplete. Thus the flow pattern information for this inclination angle was not corrected.

Fig. A-2 presents the flow pattern map corresponding to Fig. A-1 after the grouping and correction described above. Similar to vertical flow, the main flow patterns are elongated bubble flow, slug flow, churn flow and annular-mist flow.

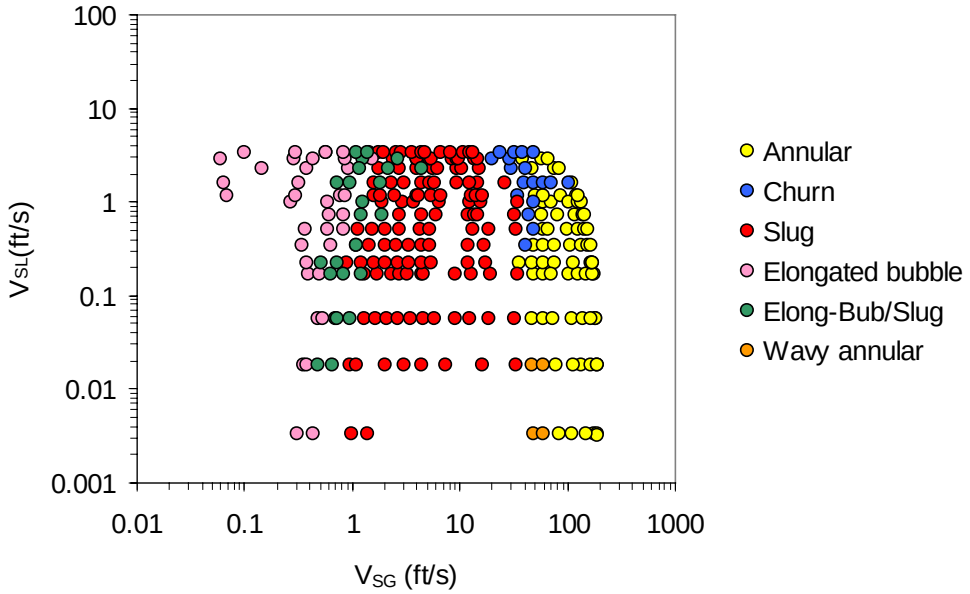


Figure A-2: Flow pattern map for Spedding & Nguyen (1976) data after correction (θ=70°, air-water, D=1.79 inch)